

AOS-CX 10.16.xxxx Monitoring Guide

8400 Switch Series



Hewlett Packard
Enterprise

Published: August 2025

Version: 1

Copyright Information

© Copyright 2025 Hewlett Packard Enterprise Development LP.

This product includes code licensed under certain open source licenses which require source compliance. The corresponding source for these components is available upon request. This offer is valid to anyone in receipt of this information and shall expire three years following the date of the final distribution of this product version by Hewlett Packard Enterprise Company. To obtain such source code, please check if the code is available in the HPE Software Center at <https://myenterpriselicense.hpe.com/cwp-ui/software> but, if not, send a written request for specific software version and product for which you want the open source code. Along with the request, please send a check or money order in the amount of US \$10.00 to:

Hewlett Packard Enterprise Company
Attn: General Counsel
WW Corporate Headquarters
1701 E Mossy Oaks Rd Spring, TX 77389
United States of America.



Notices

The information contained herein is subject to change without notice. The only warranties for Hewlett Packard Enterprise products and services are set forth in the express warranty statements accompanying such products and services. Nothing herein should be construed as constituting an additional warranty. Hewlett Packard Enterprise shall not be liable for technical or editorial errors or omissions contained herein.

Confidential computer software. Valid license from Hewlett Packard Enterprise required for possession, use, or copying. Consistent with FAR 12.211 and 12.212, Commercial Computer Software, Computer Software Documentation, and Technical Data for Commercial Items are licensed to the U.S. Government under vendor's standard commercial license.

Links to third-party websites take you outside the Hewlett Packard Enterprise website. Hewlett Packard Enterprise has no control over and is not responsible for information outside the Hewlett Packard Enterprise website.

Acknowledgment

Intel®, Itanium®, Optane™, Pentium®, Xeon®, Intel Inside®, and the Intel Inside logo are trademarks of Intel Corporation in the U.S. and other countries.

Microsoft® and Windows® are either registered trademarks or trademarks of Microsoft Corporation in the United States and/or other countries.

Adobe® and Acrobat® are trademarks of Adobe Systems Incorporated.

Java® and Oracle® are registered trademarks of Oracle and/or its affiliates.

UNIX® is a registered trademark of The Open Group.

All third-party marks are property of their respective owners.

Contents

About this document	8
Applicable products	8
Latest version available online	8
Command syntax notation conventions	8
About the examples	9
Identifying switch ports and interfaces	9
Identifying modular switch components	10
Aruba 8400 switch series member, slot, and port notation	11
Line Modules and Management Modules	11
Monitoring hardware through visual observation	13
Confirming normal operation of the switch by reading LEDs	13
Detecting if the switch is not ready for a failover event	14
Finding faulted components using the switch LEDs	14
Boot commands	16
boot fabric-module	16
boot line-module	17
boot management-module	18
boot management-module (recovery console)	19
boot set-default	20
boot system	21
show boot-history	23
External storage	27
External storage commands	27
address	27
directory	28
disable	29
enable	29
external-storage	30
password (external-storage)	31
show external-storage	32
show running-config external-storage	32
type	33
username	34
vrf	35
IP-SLA	36
IP-SLA guidelines	36
Limitations with VoIP SLAs	37
IP-SLA commands	37
http	37
https	39
icmp-echo	41
ip-sla	42
ip-sla responder	43

show ip-sla all	45
show ip-sla responder	47
show ip-sla	48
start-test	50
stop-test	50
tcp-connect	51
udp-echo	52
udp-jitter-voip	54
vrf	56
show interface	56
show interface statistics	63
Mirroring	67
Mirroring and sFlow	67
Mirror statistics	68
Classifier policies and mirroring sessions	68
Mirroring commands	69
clear mirror	69
comment	70
copy tcpdump-pcap	71
copy tshark-pcap	72
destination cpu	72
destination interface	73
destination tunnel	74
diagnostic	76
diag utilities tcpdump	77
disable	79
enable	79
mirror session	80
show mirror	81
source interface	83
source vlan	85
Monitoring a device using SNMP	88
Breakout cable support	89
Limitations with breakout cable support	89
Breakout cable support commands	89
split	89
Aruba AirWave	92
SNMP support and AirWave	92
SNMP on the switch	92
Supported features with AirWave and the AOS-CX switch	93
Configuring the AOS-CX switch to be monitored by AirWave	93
AirWave commands	94
logging	94
snmp-server community	96
snmp-server host	97
snmp-server vrf	99
snmpv3 context	99
snmpv3 user	100
Support and Other Resources	103
Accessing HPE Aruba Networking Support	103
Accessing Updates	104

Warranty Information	104
Regulatory Information	104
Documentation Feedback	104

Chapter 1

About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing HPE Aruba Networking switches on a network.

Applicable products

This document applies to the following products:

- HPE Aruba Networking 8400 Switch Series (JL366A, JL363A, JL687A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in [Support and Other Resources](#).

Command syntax notation conventions

Convention	Usage
<code>example-text</code>	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example-text	In code and screen examples, indicates text entered by a user.
Any of the following: ▪ <code><example-text></code> ▪ <code><example-text></code> ▪ <i>example-text</i> ▪ <i>example-text</i>	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: ▪ For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. ▪ For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.

Convention	Usage
[]	Brackets. Indicates that the enclosed item or items are optional.
... or ...	Ellipsis: ■ In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. ■ In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term **switch**, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME) #
```

Indicates the configuration context for a feature. For example:

```
switch(config-if) #
```

Identifies the **interface** context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>) #
```

Where **<VLAN-ID>** is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format: **member/slot/port**.

On the HPE Aruba Networking 8400 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Specifies physical location of a module in the switch chassis.
 - Management modules are on the front of the switch in slots 1/5 and 1/6.
 - Line modules are on the front of the switch in slots 1/1 through 1/4, and 1/7 through 1/10.
- *port*: Physical number of a port on a line module

For example, the logical interface **1/1/4** in software is associated with physical port 4 in slot 1 on member 1.

Identifying modular switch components

- Power supplies are on the front of the switch behind the bezel above the management modules. Power supplies are labeled in software in the format: *member/power supply*:
 - *member*: 1.
 - *power supply*: 1 to 4.
- Fans are on the rear of the switch and are labeled in software as: *member/tray/fan*:
 - *member*: 1.
 - *tray*: 1 to 4.
 - *fan*: 1 to 4.
- Fabric modules are not labeled on the switch but are labeled in software in the format: *member/module*:
 - *member*: 1.
 - *member*: 1 or 2.
- The display module on the rear of the switch is not labeled with a member or slot number.

Aruba 8400 switch series member, slot, and port notation

The software notation for describing member, slot, and port information depends on the switch hardware.

The physical interfaces on the Aruba 8400 Switch Series use the format:

member/slot/port

member

Specifies the chassis number. In this release of the software, the value of *member* is always 1.

slot

Specifies physical location in the switch chassis.

port

Specifies the physical port on the module.

The slot numbers are unique to each type of component—in contrast to being unique within a chassis.

Line Modules and Management Modules

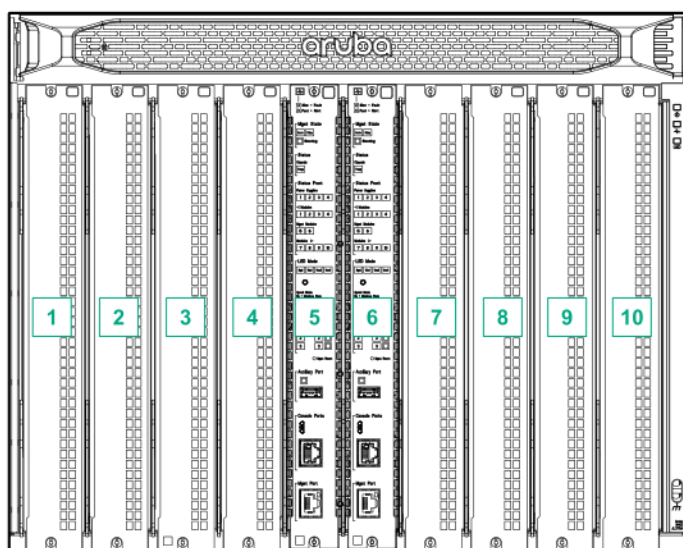
Line modules are on the front of the switch in slots 1/1 through 1/4 and 1/7 through 1/10.

The number of ports depend on the line module. Line module ports are labeled in software as port or interface, depending on the context.

For example, `interface 1/1/1` is the logical interface associated with the physical interface member 1, slot 1, port 1.

Management modules are on the front of the switch in slots 1/5 and 1/6.

Figure 1 Aruba 8400 Switch Series line module and management module slots



Power supplies

Power supplies are on the front of the switch behind the bezel above the line modules and management modules. Power supplies are labeled in software as Member/PSU: 1/1 through 1/4.

Fan trays

Fan trays are on the rear of the switch and are labeled in software as Member/Tray: 1/1 through 1/3.

Fans

Fans are on the rear of the switch in fan trays and are labeled in software as Member/Tray/Fan:

- 1/1/1 through 1/1/6
- 1/2/1 through 1/2/6
- 1/3/1 through 1/3/6

Fabric modules

Fabric modules are on the rear of the switch, behind the fan trays, in slots 1/1 through 1/3.

Rear display module

The rear display module is on the rear of the switch and is not labeled with a member or slot number.

Confirming normal operation of the switch by reading LEDs

This task describes using the switch LEDs to confirm that the switch is operating normally.



For complete information on LED behaviors for your AOS-CX switch, refer to the **Installation and Getting Started Guide** for that switch series, available for download from the [Aruba Switch Documentation](#) section of the [Aruba Hardware Documentation and Translations Portal](#).

Procedure

1. Quick check: Verify that the chassis has power and there are no fault conditions.
On the front of the switch, verify that the states of the following LEDs are On Green:
 - Power
 - Health
2. Verify that the Health LEDs of all installed line modules are On Green.
3. Verify that the Health LEDs of all installed management modules are On Green.
4. Verify that the network ports are operating normally.
 - a. On the active management module, check the Status Front section. Verify that each LED that indicates a line module is in one of the following states:
 - On Green (normal operation)
 - Off (no line module installed)
 - b. On each line module, verify that each port LED is in one of the following states:
 - On Green, Half-Bright Green, or Flickering Green (normal operation)
 - Off (no cable connected or port off by default in config)
5. Verify that the power supplies are operating normally.
 - a. On the active management module, check the Status Front section. Verify that each LED that indicates a power supply is in one of the following states:
 - On Green (normal operation)
 - Off (no power supply installed)
 - b. On each power supply, verify that LEDs are in the following states:
 - Power LED: On Green
 - Fault LED: Off
6. Verify that the rear components are operating normally by checking the Status Rear section of the active management module:
 - a. Verify that the LEDs for the fabric modules are in one of the following states:
 - On Green (normal operation)
 - Off (component not installed)
 - b. Verify that the LEDs for the fan trays and fans are On Green.

7. Verify that the standby management module is ready to take over as the active management module. On the standby management module, verify the states of the following LEDs:
 - Health LED is On Green.
 - Management state standby (Stby) LED is On Green.

Detecting if the switch is not ready for a failover event

This task describes using the switch LEDs to detect if the switch is not ready for the loss of a fabric module or for a failover from the active management module to the standby management module.



Although you can detect power supply failures by viewing the LEDs, you must use software commands to determine if the power supply redundancy is sufficient to power the chassis if a power supply fails. For complete information on LED behaviors for your AOS-CX switch, refer to the **Installation and Getting Started Guide** for that switch series, available for download from the [Aruba Switch Documentation](#) section of the [Aruba Hardware Documentation and Translations Portal](#).

Procedure

1. Detect if the standby management module is shut down.
If the standby management module is shut down, the LED states are as follows:
 - The standby management module health LED is Off.
 - The standby management state active (Actv) LED is Off.
 - The standby management state standby (Stby) LED is Off.
 - On the active management module in the Status Front Management Modules section, the LED for the standby management module is Off. For example, if the active management module is Management Module LED 5, Management Modules LED 6 is Off.
2. Detect if the standby management module is in a transient state. If the standby management module is booting, updating, or in another transient state, the LED states are as follows:
 - The standby management module health LED is Slow Flash Green when the service operating system is running or during an operating system update.
 - The standby management module Booting LED is Slow Flash Green when the AOS-CX operating system is booting.
 - The standby management state active (Actv) LED is Off.
 - The standby management state standby (Stby) LED is Off.
 - On the active management module in the Status Front Management Modules section, the LED for the standby management module is Slow Flash Green.
3. Detect if a fabric module is shut down or not present. If a fabric module is shut down or not present, the LED states are as follows:
 - On the active management module, in the Status Rear section, the LED for the fabric module is Off.
 - On the rear display module, the LED for the fabric module is Off.
 - On the fabric module, the health LED is Off. However, the fabric module is behind fan 1 and is not directly visible.

Finding faulted components using the switch LEDs

This task describes using the switch LEDs to find components that are in a fault condition.



All green LEDs—except for chassis power LEDs and the Usr1 LED—are off when the LED mode is set to Light Faults (The Usr1 LED of the LED Mode section of the active management module is On Green and the default behavior for the Usr1 LED is being used.). For complete information on LED behaviors for your AOS-CX switch, refer to the **Installation and Getting Started Guide** for that switch series, available for download from the [Aruba Switch Documentation](#) section of the [Aruba Hardware Documentation and Translations Portal](#).

Procedure

1. Find the switch that has the fault condition, which is indicated by a chassis health LED in the state of Slow Flash Orange.

The chassis health LED is located on the front of the switch and on the rear panel of the switch.

2. If you are at the back of the switch, on the rear panel, look for LEDs that are in the Slow Flash Orange state:

The Status Rear area has LEDs for power supplies, fabric modules, fan trays, and fans. The number on the LED represents the unit number of the component.

If the only LED in a state of Slow Flash Orange is the Chassis health LED, go to the front of the switch.

3. At the front of the switch, on the active management module, look for LEDs that are in the Slow Flash Orange state:
 - The Status Front area has LEDs for power supplies, line and fabric modules, and management modules. The number on the LED indicates the slot number of the component.
 - The Status Rear area has LEDs for fabric modules and fan trays, with a single LED for all the fans in the fan tray. The number on the LED represents the slot or bay number of the component.

4. Use the number indicated by the LED that is flashing to locate the slot that contains the faulted component.

The fabric modules are located behind the fan trays, and the fabric module number corresponds to the fan tray number.

5. At the front of the switch, on line modules, look for LEDs that are in the Slow Flash Orange state: Module LEDs and Port LEDs indicate faults if their states are Slow Flash Orange.

boot fabric-module

boot fabric-module <SLOT-ID>

Description

Reboots the specified fabric module.

Parameter	Description
<SLOT-ID>	Specifies the member and slot of the module in the format member/slot . For example, to specify the module in member 1 slot 3, enter 1/3 .

Usage

The **boot fabric-module** command reboots the specified fabric module. Traffic performance is affected while the module is down.

If the specified module is the only fabric module in an up state, rebooting that module stops traffic switching between line modules and the line modules power down. The line modules power up when one fabric module returns to an up state.

This command is valid for fabric modules only.

Examples

Rebooting the fabric module in slot **1/3** when auto-confirm is not enabled:

```
switch# boot fabric-module 1/3
This command will reboot the specified fabric module. Traffic performance may
be affected while the module is down. Rebooting the last fabric module will
stop traffic switching between line modules.
Do you want to continue (y/n)? y

switch#
```

Rebooting the fabric module in slot **1/1** when auto-confirm is enabled:

```
switch# boot fabric-module 1/3
This command will reboot the specified fabric module. Traffic performance may
be affected while the module is down. Rebooting the last fabric module will
stop traffic switching between line modules.
Do you want to continue (y/n) y (auto-confirm)

switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	Manager (#)	Administrators or local user group members with execution rights for this command.

boot line-module

`boot line-module <SLOT-ID>`

Description

Reboots the specified line module.

Parameter	Description
<code><SLOT-ID></code>	Specifies the member and slot of the module in the format member/slot . For example, to specify the module in member 1 slot 3, enter 1/3 .

Usage

Reboots the specified line module. Any traffic for the switch passing through the affected module (SSH, TELNET, and SNMP) is interrupted. It can take up to 2 minutes to reboot the module. During that time, you can monitor progress by viewing the event log.

This command is valid for line modules only.

Examples

Reloading the module in slot 1/1:

```
switch# boot line-module 1/1
This command will reboot the specified line module and interfaces on this
module will not send or receive packets while the module is down. Any
traffic passing through the line module will be interrupted. Management
sessions connected through the line module will be affected. It might take
up to 2 minutes to complete rebooting the module. During that time, you can
monitor progress by viewing the event log.
Do you want to continue (y/n)? y
switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	Manager (#)	Administrators or local user group members with execution rights for this command.

boot management-module

`boot management-module {active | standby | <SLOT-ID>}`

Description

Reboots the specified management module. Choose the management module to reboot by role (active or standby) or by slot number.

Parameter	Description
<i>active</i>	Selects the active management module.
<i>standby</i>	Selects the standby management module.
<i><SLOT-ID></i>	Specifies the member and slot of the management module in the format member/slot . For example, to specify the module in member 1 slot 5, enter 1/5 .

Usage

This command reboots a single management module in a chassis. Choose the management module to reboot by role (active or standby) or by slot number.

You can use the **show images** command to show information about the primary and secondary system images.

If you reboot the active management module and the standby management module is available, the active management module reboots and the standby management module becomes the active management module.

If you reboot the active management module and the standby management module is not available, you are warned, you are prompted to save the configuration, and you are prompted to confirm the operation.

If you reboot the standby management module, the standby management module reboots and remains the standby management module.

If you attempt to reboot a management module that is not available, the **boot** command is aborted.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the **boot** command is aborted.



Hewlett Packard Enterprise recommends that you use the **boot management-module** command instead of pressing the module reset button to reboot a management module because if you are rebooting the only available management module, the **boot management-module** command enables you to save the configuration, cancel the reboot, or both.

Examples

Rebooting the active management module when the standby management module is available:

```
switch# boot management-module active
The management-module in slot 1/5 is going down for reboot now.
```

Rebooting the active management module when the standby management module is not available:

```
switch# boot management-module 1/5
The management module in slot 1/5 is currently active and no
standby management module was found.
This will reboot the entire switch.

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	Manager (#)	Administrators or local user group members with execution rights for this command.

boot management-module (recovery console)

boot management-module {local|remote}

Description

Reboots the specified management module by specified location (local or remote).

Parameter	Description
<local>	Reboots the local management module.
<remote>	Reboots the remote management module.

Usage

This command reboots a single management module in a chassis. Choose the management module to reboot by role (active or standby) or by slot number.

You can use the **show images** command to show information about the primary and secondary system images.

If you reboot the active management module and the standby management module is available, the active management module reboots and the standby management module becomes the active management module.

If you reboot the active management module and the standby management module is not available, you are warned, you are prompted to save the configuration, and you are prompted to confirm the operation.

If you reboot the standby management module, the standby management module reboots and remains the standby management module.

If you attempt to reboot a management module that is not available, the **boot** command is aborted.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the **boot** command is aborted.



Hewlett Packard Enterprise recommends that you use the **boot management-module** command instead of pressing the module reset button to reboot a management module because if you are rebooting the only available management module, the **boot management-module** command enables you to save the configuration, cancel the reboot, or both.

Examples

Booting a remote management module:

```
switch# boot management-module remote
There is no other management module installed.
Aborting.
switch#
```

Command History

Release	Modification
10.12	Command introduced.

Command Information

Platforms	Command context	Authority
8400	Manager (#)	Administrators or local user group members with execution rights for this command.

boot set-default

```
boot set-default {primary | secondary}
```

Description

Sets the default operating system image to use when the system is booted. Changes to this setting may impact other features, such as the job scheduler feature configured via the **reload at** or **reload after** commands.

Parameter	Description
primary	Selects the primary network operating system image.
secondary	Selects the secondary network operating system image.

Example

Selecting the primary image as the default boot image:

```
switch# boot set-default primary
Default boot image set to primary.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

boot system

boot system [primary | secondary | serviceos]

Description

Reboots all modules on the switch. By default, the configured default operating system image is used. Optional parameters enable you to specify which system image to use for the reboot operation and for future reboot operations.

Parameter	Description
primary	Selects the primary operating system image for this reboot and sets the configured default operating system image to primary for future reboots.
secondary	Selects the secondary operating system image for this reboot and sets the configured default operating system image to secondary for future reboots.
serviceos	Selects the service operating system for this reboot. Does not change the configured default operating system image. The service operating system acts as a standalone bootloader and recovery OS for switches running the AOS-CX operating system and is used in rare cases when troubleshooting a switch.

Usage

This command reboots the entire system. If you do not select one of the optional parameters, the system reboots from the configured default boot image.

You can use the **show images** command to show information about the primary and secondary system images.

Choosing one of the optional parameters affects the setting for the default boot image:

- If you select the **primary** or **secondary** optional parameter, that image becomes the configured default boot image for future system reboots. The command fails if the switch is not able to set the operating system image to the image you selected.

You can use the **boot set-default** command to change the configured default operating system image.

- If you select **serviceos** as the optional parameter, the configured default boot image remains the same, and the system reboots all management modules with the service operating system.

If the configuration of the switch has changed since the last reboot, when you execute the **boot system** command you are prompted to save the configuration and you are prompted to confirm the reboot operation.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the **boot system** command is aborted.

Examples

Rebooting the system from the configured default operating system image:

```
switch# boot system
Do you want to save the current configuration (y/n)? y
The running configuration was saved to the startup configuration.

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
The system is going down for reboot.
```

Rebooting the system from the secondary operating system image, setting the secondary operating system image as the configured default boot image:

```
switch# boot system secondary
Default boot image set to secondary.

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Canceling a system reboot:

```
switch# boot system

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
```

```
until the process is complete.  
Continue (y/n)? n  
Reboot aborted.  
switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show boot-history

```
show boot-history [all|{vsf member <1-10>}]
```

Description

Shows boot history information. When no parameters are specified, shows the most recent information about the current boot operation, and the three previous boot operations for the switch. When the **all** parameter is specified, the output of this command shows the boot information for the active management module.

For switches that support line modules (such as 8400 switch series) including the **all** parameter displays information for the active management module and all available line modules.



To view boot-history on a standby, the command must be sent on the conductor console.

Parameter	Description
all	Optional. Shows boot information for the active management module. For switches that support line modules, including this parameter displays information for and all available line modules.
vsf member <1-10>	Optional. Display boot history for the specified VSF member

Usage

This command displays the boot-index, boot-ID, and up time in seconds for the current boot. If there is a previous boot, it displays boot-index, boot-ID, reboot time (based on the time zone configured in the system) and reboot reasons. Previous boot information is displayed in reverse chronological order.

The output of this command includes the following information:

Parameter	Description
Index	The position of the boot in the history file. Range: 0 to 3.
Boot ID	A unique ID for the boot . A system-generated 128-bit string.
Current Boot, up for <time>	For the current boot, the show boot-history command shows the number of seconds the module has been running on the current software.
<Timestamp>: boot reason	For previous boot operations, the show boot-history command shows the time at which the operation occurred and the reason for the boot. The reason for the boot is one of the following values: ▪ <DAEMON-NAME> crash: The daemon identified by <DAEMON-NAME> caused the module to boot. ▪ Kernel crash: The operating system software associated with the module caused the module to boot. ▪ Uncontrolled reboot: The reason for the reboot is not known. ▪ Reboot requested through database: The reboot occurred because of a request made through the CLI or other API.

Examples

Showing the boot history of the active management module:

```
switch# show boot-history
Management module
=====

Index : 2
Boot ID : c34a2c2499004a02bbeeff4992e1fdbd
Current Boot, up for 1 days 13 hrs 13 mins 27 secs

Index : 1
Boot ID : bfba9bc486304e57904ac717a0ccbdcd
02 Sep 23 02:55:33 : CPU request reset with 0x20201, Version: FL.10.14.0000-1619-
ga9ec1805bd442~dirty
02 Sep 23 02:55:33 : Switch boot count is 2

Index : 0
Boot ID : a88a71b7ca9a4574af7e3b811ddfdc7e
02 Sep 23 02:49:26 : Reboot requested by user, Version: FL.10.14.0000-1619-
ga9ec1805bd442~dirty
02 Sep 23 02:50:02 : Switch boot count is 1

Index : 3
Boot ID : f00ba10c8c44457f83fee303d014a89a
25 Aug 23 10:27:42 : Power on reset with 0x1, Version: FL.10.14.0000-1465-
g9df95249d06b0~dirty
25 Aug 23 10:28:18 : Switch boot count is 3
25 Aug 23 10:29:02 : Primary overtemperature fault detected with 0x2 in PSU 1/1
```


Showing the boot history of the active management module and all line modules:

```
switch#
Management module
=====

Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs

Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database

Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database

Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database

Line module 1/1
=====
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...

Management module
=====

Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs

Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database

Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database

Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database

Line module 1/1
=====
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

The switch has limited capacity to store data, collected by switch features and protocols. You can provide virtually unlimited storage capacity by adding user-supplied external storage volumes. Supported volume types and storage protocols include: NFSv3, NFSv4, and SCP (sshfs).

One application of external storage is the saving and restoring of DHCP lease files over SCP or NFS network attached storage systems. SCP file system protocol uses a user mode process to emulate a network file system. The key advantage is packet level encryption and simple configuration. The key disadvantage is slow performance.

You can set up external storage volume credentials and then enable it. A storage management process acts on your requests by enabling the storage volume using the requested storage protocol. You can disable the external storage volume or set it up but leave it disable.

The feature maintains storage volume state. The states are: **disabled** (down), **connecting** (establishing connection), **operational** (up), and **unaccessible** (unavailable).

If a storage volume is unavailable, the system attempts to reconnect periodically. Multiple volumes could connect concurrently. If one connection times out the others can connect immediately.

The system supports server connection through data and management ports.

Data port support requires server IP address on a default VRF.

Once a storage volume is enabled, applications can use the volume to store retrieve and delete files and directories.

External storage commands

address

```
address {<IPV4-ADDR> | <IPV6-ADDR> | hostname <HOSTNAME>}  
no address {<IPV4-ADDR> | <IPV6-ADDR> | hostname <HOSTNAME>}
```

Description

Specifies the NAS IP address or hostname.

The **no** form of this command deletes an IP address or hostname.

Parameter	Description
<IPV4-ADDR>	Specifies the NAS server IPv4 address, Global.
<IPV6-ADDR>	Specifies the IPv6 address of the NAS server.
<HOSTNAME>	Specifies the hostname of the NAS server. String.

Examples

Creating the logfiles storage volume with IP address 10.1.1.1:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# address 10.1.1.1
```

Deleting an external storage volume named logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no address 10.1.1.1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	config-external-storage-<VOLUME-NAME>	Administrators or local user group members with execution rights for this command.

directory

```
directory <DIRECTORY-NAME>
no directory <DIRECTORY-NAME>
```

Description

Selects an existing directory on the external storage volume.

The **no** form of this command clears a directory of an external storage volume.

Parameter	Description
<DIRECTORY-NAME>	Specifies the external storage directory for mapping the volume.

Examples

Creating a volume named logfiles that is mapped under /home on the server:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# directory /home
```

Clearing the directory /home:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no directory /home
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	config-external-storage-<VOLUME-NAME>	

disable

disable
no disable

Description

Disables the external storage volume.

The **no** form of this command enables the external storage volume. This is identical to the `enable` command.

Examples

Disabling a volume named logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# disable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	config-external-storage-<VOLUME-NAME>	

enable

enable
no enable

Description

Enables the external storage volume.

The **no** form of this command disables the external storage volume. This is identical to the `disable` command.

Examples

Creating and then enabling a volume named logfiles:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# enable
```

Disables the external storage volume:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# disable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	config-external-storage- <i><VOLUME-NAME></i>	

external-storage

```
external-storage <VOLUME-NAME>  
no external-storage <VOLUME-NAME>
```

Description

Creates or updates an external storage volume.

The **no** form of this command deletes an external storage volume.

Examples

Creating the logfiles storage volume:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)#
```

Deleting the logfiles storage volume:

```
switch(config)# no external-storage logfiles
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	config	Administrators or local user group members with execution rights for this command.

password (external-storage)

```
password [{plaintext | ciphertext} <PASSWORD>]
no password {plaintext | ciphertext} <PASSWORD>
```

Description

Sets the password for network attached storage server login.

The **no** form of this command clears the password for network attached storage server login.

Parameter	Description
{ciphertext plaintext}	Selects the password format.
<PASSWORD>	Specifies the password. NOTE: When the password is not provided on the command line, plaintext password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.

Examples

Creating a volume named logfiles with password Xj#9:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# password plaintext Xj#9
```

Creating a volume named bak1 with a prompted plaintext password:

```
switch(config)# external-storage bak1
switch(config-external-storage-bak1)# password
Enter the NAS server password: *****
Re-Enter the NAS server password: *****
```

Clearing the password for volume logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no password plaintext Xj#9
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	config-external-storage-<VOLUME-NAME>	Administrators or local user group members with execution rights for this command.

show external-storage

show external-storage [<VOLUME-NAME>]

Description

Shows external storage configuration and state for all volumes or for a specified volume.

Parameter	Description
<VOLUME-NAME>	Specifies the external storage volume name that the show command will use.

Examples

```
switch# show external-storage
```

```
-----
--
      Address      VRF      Username      Type      Directory      State
-----
--
nfsvol    10.1.1.1      nas      ---           NFSv3      /home
operational
nfsfiles  20.1.1.1      nas      netstorage    NFSv4      /netstor      disabled
scpdev    nasserver     nas      scpstor       SCP        /scp
unaccessible
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show running-config external-storage

show running-config external-storage

Description

Shows the running configuration of the external storage.

Examples


```
switch# show running-config external-storage
```

```
external-storage nfsvol
  address 10.1.1.1
  vrf     nas
  type    nfsv4
  directoty /home
  enable
external-storage scpdev
  address 30.1.1.1
  vrf     nas
  username switchuser
  password ciphertext xxx
  type    scp
  directoty /home
  enable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

type

```
type {nfsv3 | nfsv4 | scp}
no type {nfsv3 | nfsv4 | scp}
```

Description

Sets the network attached storage access type for reaching the external storage volume. The **no** form of this command deletes an external storage volume.

Parameter	Description
nfsv3	Specifies the NFSv3 network access protocol.
nfsv4	Specifies the NFSv4 network access protocol.
scp	Specifies the SCP network access protocol.

Examples

Creating the logfiles volume using NFSV4:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# type nfsv4
```

Clearing the external storage access type:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# no type nfsv4
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	config-external-storage- <i><VOLUME-NAME></i>	Administrators or local user group members with execution rights for this command.

username

```
username <USER-NAME>  
no username <USER-NAME>
```

Description

Sets the username for logging in to a network attached storage server.

The **no** form of this command clears a username.

Parameter	Description
<i><USER-NAME></i>	Specifies the username.

Examples

Creating a volume named logfiles with the user name nassuser:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# username nasuser
```

Clearing the user name nasuser from accessing the logfiles volume:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# no username nasuser
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	config-external-storage-<VOLUME-NAME>	Administrators or local user group members with execution rights for this command.

vrf

```
vrf <VRF-NAME>
no vrf <VRF-NAME>
```

Description

Setting a VRF to reach network attached storage.

The **no** form of this command clears access of a VRF to network attached storage.

Parameter	Description
<VRF-NAME>	Specifies the VRF name.

Examples

Creating the logfiles volume and setting a VRF named nas to access the network attached storage:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# vrf nas
```

Clearing access of a VRF named nas to the network attached storage:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no vrf nas
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	config-external-storage-<VOLUME-NAME>	Administrators or local user group members with execution rights for this command.

The IP Service Level Agreement (IP-SLA) is a feature that enables the measuring of network performance between two nodes in a network for different service level agreement parameters such as round-trip time (RTT), one-way delay, jitter, reachability, packet loss, and voice quality scores. These two nodes can span across area in access, distribution or core inside a LAN as well as across WAN between core to core or core to Data Centre switches. This feature helps you measure the SLA for different protocols or applications such as UDP echo, UDP jitter (for voice and video), TCP connect, HTTP, and ICMP echo. This guide provides details for managing and monitoring different types of IP-SLAs.

IP-SLA guidelines

- AOS-CX supports only SLA configuration through CLI and thresholds can be configured using NAE agents using WebUI/REST.
- AOS-CX supports only forever tests. On-demand tests are not supported.
- Maximum sessions: IP-SLA source 500, IP-SLA responder 500.
- NAE can effectively monitor a maximum of 300 parameters, reducing the maximum supported session by 300.
- NAE supports only syslog.
- NAE agents must be triggered for each IP-SLA test on every switch.
- If multiple IP addresses are received for a DNS query, DNS works with the first resolved IP.
- When the DNS server IP is not explicitly configured, the system automatically uses the first DNS server available in its default configuration.
- The source interface/IP option is not applicable for SLAs configured on 'mgmt' VRF, as it has only one interface.
- A system time change because of NTP or a manual change causes an incorrect calculation.
- There is no interoperability of UDP echo SLA between AOS-CX and FlexFabric switches.
- Source IP and source port combination must be unique across SLA sessions in a same switch.
- Do not use the same source port across the source and responder sessions in a switch.
- The configuration of history results is limited to a maximum of 8 IP-SLA sessions. This means that you can enable and store historical performance data, such as response times and availability, for up to 8 individual IP SLA sessions at any given time.
- NTP synchronization is a must for SLA types involving one-way delay such as UDP jitter VoIP.
- It is mandatory to set default CoPP to the maximum value when UDP jitter SLA is enabled. Otherwise, 100% packet loss can be seen and UDP jitter SLA probes will result in failure:

```
copp-policy default
  class hypertext priority 6 rate 50000 burst 64
  default-class priority 6 rate 99999 burst 9999
```

Limitations with VoIP SLAs

- A maximum of 80 concurrent VoIP SLAs can be scheduled in a 20 second slot.
- A single VoIP probe takes 20 seconds to complete.
- The default and minimum probe interval for VoIP SLA is 120 seconds.
- SLAs scheduled in the same slot, periodically sends 1000 probe packets for 120 seconds in 20 second intervals.
- Default 120 second probe interval is divided in to 6 slots of 20 seconds to avoid synchronization of all configured VoIP SLAs sending probes at the same time.
- SLAs started at the same time exceeding the concurrent limit of 80 must wait for the next 20 second VoIP slot to open before moving to 'running' state.
- The maximum number of VoIP SLAs supported is 80 X 6 slots = 480 SLAs.
- SLAs exceeding 480 will continue to remain in the 'waiting for VoIP slot' until any slot is freed by stopping the running SLA.
- To avoid high RTT, a single switch with more than 20 SLAs should not have single responder SLA.
- When IP is received dynamically (e.g. using DHCP) for interfaces other than management interface, IPSLA source or responder has to be configured only using interface name.

IP-SLA commands

http

```
http {get | raw} <URL> [history-interval <HISTORY-INTERVAL>] [cache disable] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}] [probe-interval <PROBE-INTERVAL>] [proxy <PROXY-URL>] [source {<IPV4-ADDR>|<IPV6-ADDR>|IFNAME}] [source-port <SOURCE-PORT-NUM>] [version <VERSION-NUMBER>] [http-raw-request <RAW-PAYLOAD>]
```

```
no http {get | raw} <URL> [history-interval <HISTORY-INTERVAL>] [cache disable] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}] [probe-interval <PROBE-INTERVAL>] [proxy <PROXY-URL>] [source {<IPV4-ADDR>|<IPV6-ADDR>|IFNAME}] [source-port <SOURCE-PORT-NUM>] [version <VERSION-NUMBER>] [http-raw-request <RAW-PAYLOAD>]
```

Description

Configures HTTP as the IP-SLA test mechanism. Requires destination URL and type of HTTP request (raw/get).

The **no** version of this command disables HTTP as the IP-SLA test mechanism.

Parameter	Description
{get raw}	Selects HTTP request type as get or raw where the system will generate or provide HTTP payload.
<URL>	Specifies HTTP URL address of syntax http://<HOST NAME/IP-ADDRESS>:<PORT>/<PATH>.
history-interval <HISTORY-INTERVAL>	Configures the history interval for the IP-SLA. Set the history

Parameter	Description
	interval to minimum of two times the probe-interval for the SLA. Range: 60 to 7200.
cache disable	Selects cache option for the HTTP server. By default the option is enabled.
name-server {<IPV4-ADDR-DNS-SERVER> <IPV6-ADDR-DNS-SERVER>}	Specifies the DNS server for destination hostname resolution.
probe-interval <PROBE-INTERVAL>	Specifies the probe interval in seconds. Range: 30 to 604800.
proxy <PROXY-URL>	Specifies the probe interval in seconds. Range: 30 to 604800.
source {<IPV4-ADDR> <IPV6-ADDR> IFNAME}	Selects the source, either an IPv4 address, an IPv6 address, or hostname for SLA probes.
source-port <SOURCE-PORT-NUM>	Specifies the value of the source port for the IP-SLA probes.
version <VERSION-NUMBER>	Specifies the source interface to use for sending IP-SLA probes.
http-raw-request <RAW-PAYLOAD>	Specifies the HTTP raw request. String.

Examples

Configuring HTTP get with parameters, including history interval:

```
switch(config)# ip-sla 1
switch(config-ipsla-1)# http get http://device.arubanetworks.com/root/home.html
history-interval 120 cache disable name-server 10.10.10.2 probe-interval 30
```

Configuring HTTP raw with parameters:

```
switch(config-ipsla-1)# http raw http://2.2.2.2 http-raw-request "GET
/en/US/hmpgs/index.html HTTP/1.0\r\n\r\n"
```

Disabling HTTP get with parameters:

```
switch(config-ipsla-1)# no http get http://device.example.com/root/home.html name-
server 10.10.10.2 history-interval 120
```

Disabling HTTP raw with parameters:

```
switch(config-ipsla-1)# no http raw http://device.example.com/root/home.html http-raw-request "GET /en/US/hmpgs/index.html HTTP/1.0\r\n\r\n"
```

Command History

Release	Modification
10.16.1000	Added new parameter history-interval . Also, IPv6 addresses can be used.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	config-ip-sla-<IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

https

```
https {get | raw} <URL> [history-interval <HISTORY-INTERVAL>] [cache disable] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}] [probe-interval <PROBE-INTERVAL>] [proxy <PROXY-URL>] [source {<IPV4-ADDR>|<IPV6-ADDR>|IFNAME}] [source-port <SOURCE-PORT-NUM>] [version <VERSION-NUMBER>] [http-raw-request <RAW-PAYLOAD>]
```

```
no https {get | raw} <URL> [history-interval <HISTORY-INTERVAL>] [cache disable] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}] [probe-interval <PROBE-INTERVAL>] [proxy <PROXY-URL>] [source {<IPV4-ADDR>|<IPV6-ADDR>|IFNAME}] [source-port <SOURCE-PORT-NUM>] [version <VERSION-NUMBER>] [http-raw-request <RAW-PAYLOAD>]
```

Description

Configures HTTPS as the IP-SLA test mechanism. Requires destination URL and type of HTTPS request (get/raw).

The **no** form of this command removes the configuration.



For HTTPS IP-SLA sessions, it is not required to install a certificate on the switch.

Parameter	Description
{get raw}	Selects HTTPS request type as get or raw where the system will generate or provide HTTPS payload.
<URL>	Specifies HTTPS URL address of syntax. https://<HOST NAME/IP-ADDRESS>:<PORT>/<PATH>.
history-interval <HISTORY-INTERVAL>	Configures the history interval

Parameter	Description
	for the IP-SLA. Set the history interval to minimum of two times the probe-interval for the SLA. Range: 60 to 7200.
cache disable	Selects cache option for the HTTPS server. By default the option is enabled.
name-server {<IPv4-ADDR-DNS-SERVER> <IPv6-ADDR-DNS-SERVER>}	Specifies the IPv4 address of DNS server.
probe-interval <PROBE-INTERVAL>	Specifies the probe interval in seconds. Range: 30 to 604800.
proxy <PROXY-URL>	Specifies the probe interval in seconds. Range: 30 to 604800.
source {<IPv4-ADDR> <IPv6-ADDR> IFNAME}	Selects the source, either an IPv4 address, an IPv6 address, or hostname for SLA probes.
source-port <SOURCE-PORT-NUM>	Specifies the value of the source port for the IP-SLA probes.
version <VERSION-NUMBER>	Specifies the source interface to use for sending IP-SLA probes.
https-raw-request <RAW-PAYLOAD>	Specifies the HTTPS raw request. String.

Examples

Configuring HTTPS get with parameters:

```
switch(config-ipsla-1)# https get https://device.arubanetworks.com/root/home.html
```

Configuring HTTPS raw with parameters:

```
switch(config-ipsla-1)# https raw https://device.arubanetworks.com/root/home.html
raw-request "GET /en/US/hmpgs/index.html"
```

Removing the HTTPS raw:

```
switch(config-ipsla-1)# no https raw
https://device.arubanetworks.com/root/home.html raw-request "GET
/en/US/hmpgs/index.html"
```

Command History

Release	Modification
10.16.1000	Added new parameters history-interval . Also, IPv6 addresses can be used.
10.12.1000	Command introduced.

Command Information

Platforms	Command context	Authority
8400	config-ip-sla-<IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

icmp-echo

```
icmp-echo {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} [history-interval <HISTORY-INTERVAL>] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>} [payload-size <PAYLOAD-SIZE>] [probe-interval <PROBE-INTERVAL>] [source {<SOURCE-IPV4-ADDR>|<SOURCE-IPV6-ADDR>|<IFNAME>}]] [timeout <TIMEOUT>] [tos <TYPE-OF-SERVICE>]
```

```
no icmp-echo {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} [history-interval <HISTORY-INTERVAL>] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>} [payload-size <PAYLOAD-SIZE>] [probe-interval <PROBE-INTERVAL>] [source {<SOURCE-IPV4-ADDR>|<SOURCE-IPV6-ADDR>|<IFNAME>}]] [timeout <TIMEOUT>] [tos <TYPE-OF-SERVICE>]
```

Description

Configures ICMP echo as the IP-SLA test mechanism. Requires destination address for the IP-SLA test. The **no** form of this command disables the ICMP echo as the IP-SLA test mechanism.

Parameter	Description
{<DEST-IPV4-ADDR> <DEST-IPV6-ADDR> <HOSTNAME>}	Selects the destination, either an IPv4 address, an IPv6 address, or hostname, for the IP-SLA.
history-interval <HISTORY-INTERVAL>	Specifies the history interval for the IP-SLA. Set the history interval to minimum of two times the probe-interval for the SLA. Range: 10 to 7200.
name-server {<IPV4-ADDR-DNS-SERVER> <IPV6-ADDR-DNS-SERVER>}	Specifies the DNS server for destination hostname resolution.
payload-size <PAYLOAD-SIZE>	Specifies the payload size of an SLA probe. Range: 0 to 1440.
probe-interval <PROBE-INTERVAL>	Specifies the probe interval in seconds. Range: 5 to 604800.
source {<SOURCE-IPV4-ADDR> <SOURCE-IPV6-ADDR> <IFNAME>}	Selects the source IPv4 or IPv6 address for SLA probes or the

Parameter	Description
	source interface to use for sending IP-SLA probes.
<code>timeout <TIMEOUT></code>	Specifies the interval before a probe is timed out. Range: 5 to 604800.
<code>tos <TYPE-OF-SERVICE></code>	Specifies the type of service value to be used in probe packets. Range: 0 to 255.

Examples

Configuring icmp-echo:

```
switch(config)# ip-sla test
switch(config-ip-sla-test)# icmp-echo 2.2.2.2 name-server 4.4.4.4 source 3.3.3.3
```

Configuring ICMP echo with several parameters, including history interval and timeout:

```
switch(config)# ip-sla test
switch(config-ip-sla-test)# icmp-echo 2.2.2.2 history-interval 160 name-server
4.4.4.4 payload-size 400 probe-interval 80 source 3.3.3.3 timeout 20 tos 255
```

Command History

Release	Modification
10.16.1000	Added two new parameters history-interval and timeout . Also, IPv6 addresses can be used.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
8400	<code>config-ip-sla-<IP-SLA-NAME></code>	Administrators or local user group members with execution rights for this command.

ip-sla

```
ip-sla <IP-SLA-NAME>
no ip-sla <IP-SLA-NAME>
```

Description

Creates an IP Service Level Agreement (SLA) profile and switches to the **config-ip-sla** context.

The **no** form of this command deletes an IP-SLA profile. By default, all profile use the default VRF (default).

Parameter	Description
<i><IP-SLA-NAME></i>	Specifies an IP-SLA profile name. Length: 1 to 64 characters.

Examples

Creating an IP-SLA:

```
switch(config)# ip-sla 1
switch(config-ip-sla-1)#
```

Deleting an IP-SLA:

```
switch(config)# no ip-sla 1
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	config	Administrators or local user group members with execution rights for this command.

ip-sla responder

```
ip-sla responder <SLA-NAME> (udp-echo | tcp-connect | udp-jitter-voip) [<PORT-NUM>]
[source {<SOURCE-IPV4-ADDR>|<SOURCE-IPV6-ADDR>|<IFNAME>}] [vrf <VRF-NAME>] [ipv6]
```

```
no ip-sla responder <SLA-NAME> (udp-echo | tcp-connect | udp-jitter-voip) [<PORT-NUM>]
[source {<SOURCE-IPV4-ADDR>|<SOURCE-IPV6-ADDR>|<IFNAME>}] [vrf <VRF-NAME>] [ipv6]
```

Description

Selects the IP-SLA responder. The responder can be configured for udp-echo, tcp-connect, udp-jitter-voip type. It requires the SLA name, SLA type, and port number as arguments.

The **no** form of this command removes the IP-SLA responder.

Parameter	Description
<i><SLA-NAME></i>	Specifies the IP-SLA responder name. Length: 1 to 64 characters.
udp-echo	Enables responder for udp-echo probes.
tcp-connect	Selects TCP connect as the IP-SLA

Parameter	Description
	test mechanism.
udp-jitter-voip	Selects VOIP jitter as the IP-SLA test mechanism.
<PORT-NUM>	Specifies the port number to listen for IP-SLA probes. Range: 1 to 65535.
source {<SOURCE-IPV4-ADDR> <SOURCE-IPV6-ADDR> <IFNAME>}	Selects the source IPv4 or IPv6 address for SLA probes or the source interface to use for sending IP-SLA probes.
vrf <VRF-NAME>	Specifies the name of the VRF to use.
ipv6	Configures IPv6 responder. This keyword is required if an IPv6 address is being used by the source interface or VRF. By default, it will be considered as an IPv4 address.

Usage

The IPv6 keyword is required if an IPv6 address is being used by the source interface or VRF. Otherwise, by default, it will be considered as an IPv4 address.

Examples

Configuring IP-SLA responder for udp-echo:

```
switch(config)# ip-sla responder SLA1 udp-echo 8000 source 2.2.2.2
```

Configuring IP-SLA responder with IPv6:

```
switch(config)# ip-sla responder SLA1 udp-echo 8000 source 1/1/1 ipv6
```

Configuring IP-SLA responder for udp-jitter-voip:

```
switch(config)# ip-sla responder SLA1 udp-jitter-voip 1025 vrf <VRF>
```

Disabling IP-SLA responder:

```
switch(config)# no ip-sla responder SLA1 udp-echo 8000 source 2.2.2.2
```

Command History

Release	Modification
10.15	Added ipv6 parameter.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	config	Administrators or local user group members with execution rights for this command.

show ip-sla all

show ip-sla all

Description

Shows all ip-sla source configuration and status.

Examples

Showing results for ip-sla all:

```
switch# show ip-sla all
SLA Name           : 2 (non-persistent)
Status             : running
SLA Type           : udp-echo
VRF                : default
Source IP          :
Source Interface    :
Domain Name Server :
Payload Size       : 28
TOS                : 0
Probe Interval(seconds) : 60
History Interval(seconds) : 0
Timeout Interval(seconds) : 45

IP-SLA session status
IP-SLA Name         : 2 (non-persistent)
IP-SLA Type         : udp-echo
Destination Host Name/IP Address : 10.1.1.2
Destination Port     : 8888
Source IP Address/IFName :
Source Port         :
Status              : running

IP-SLA Session Cumulative Counters
Total Probes Transmitted : 10
Probes Timed-out         : 10
Bind Error               : 0
Destination Address Unreachable : 0
DNS Resolution Failures  : 0
Reception Error          : 0
Transmission Error       : 0
Operational Status       : down

IP-SLA Latest Probe Results
```

```

Last Probe Time           :
Packets Sent              : 1
Packets Received          : 0
Packet Loss in Test       : 100%

Minimum RTT(ms)           :
Maximum RTT(ms)           :
Average RTT(ms)           :
DNS RTT(ms)              :

-----

SLA Name                   : echo-udp-sess2 (non-persistent)
Status                     : running
SLA Type                   : udp-echo
VRF                        : default
Source IP                  :
Source Interface           :
Domain Name Server         :
Payload Size               : 28
TOS                        : 0
Probe Interval(seconds)    : 60
History Interval(seconds)  : 0
Timeout Interval(seconds)  : 45

IP-SLA session status
IP-SLA Name                : echo-udp-sess2 (non-persistent)
IP-SLA Type                : udp-echo
Destination Host Name/IP Address : 100.1.1.2
Destination Port           : 8888
Source IP Address/IFName   :
Source Port                :
Status                     : running

IP-SLA Session Cumulative Counters
Total Probes Transmitted   : 10
Probes Timed-out          : 0
Bind Error                 : 0
Destination Address Unreachable : 0
DNS Resolution Failures    : 4
Reception Error           : 0
Transmission Error        : 0
Operational Status        : Up

IP-SLA Latest Probe Results
Last Probe Time           :
Packets Sent              : 1
Packets Received          : 1
Packet Loss in Test       : 0.0000%

Minimum RTT(ms)           :
Maximum RTT(ms)           :
Average RTT(ms)           :
DNS RTT(ms)              :

-----

```

Showing results for non-configured ip-sla all:

```

switch# show ip-sla all
IPSLA source is not configured

```

Command History

Release	Modification
10.16.1000	Updated to display history interval .
10.12.1000	Updated to display https as an IP-SLA type.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
8400	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show ip-sla responder

show ip-sla responder <SLA-NAME> [initiator {<SOURCE-IPV4-ADDR>|<SOURCE-IPV6-ADDR>}] [<SOURCE-PORT-NUM>] [results]

Description

Shows the given IP-SLA responder configuration and operation status.

Parameter	Description
<SLA-NAME>	Specifies the SLA name.
initiator {<SOURCE-IPV4-ADDR> <SOURCE-IPV6-ADDR>}	Selects the source IPv4 or IPv6 address for SLA probes to use.
<SOURCE-PORT-NUM>	Configures the source port for the IP-SLA test. Range: 1 to 65535.
results	Displays the statistics for a given source IP and port.

Examples

Showing IP-SLA responder configuration:

```
switch# show ip-sla responder SLA3

SLA Name       : SLA3
IP-SLA Type    : Udp-echo
VRF            : Default
Responder Port  : 8000
Responder IP    : 2.2.2.3
Responder Interface : 1/1/1
Responder Status : Running
```

Showing IP-SLA responder with initiator and results parameters:

```
switch# show ip-sla responder SLA1 initiator 2.2.2.1 8000 results

IP-SLA Type      : Udp-echo
VRF Name         : Default
Source IP        : 2.2.2.1
Source Port      : 8000
Responder Port   : 8888
Responder IP     : 2.2.2.3
Responder Interface :
Responder Status : Running
Packets Received : 2
Packets Sent    : 2
```

Command History

Release	Modification
10.16.1000	Added new parameters: initiator and results .
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
8400	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show ip-sla

```
show ip-sla <SLA-NAME> [{results | history-results}]
```

Description

Shows the given IP-SLA source configuration and status.

Parameter	Description
<SLA-NAME>	Specifies the SLA name.
results	Displays the statistics calculated for an SLA type.
history-results	Displays the history statistics calculated for the SLA ID.

Examples

Showing results for ip-sla:

```
switch# show ip-sla xyz results
IP-SLA session status
IP-SLA Name           : xyz
IP-SLA Type           : tcp-connect
Destination Host Name/IP Address: 2.2.2.1
Destination Port      : 8888
Source IP Address/IFName : 2.2.2.2
```



```

Source Port           : 5555
Status                : running

IP-SLA session cumulative counters
Total Probes Transmitted : 1
Probes Timed-out        : 0
Bind Error              : 0
Destination Address Unreachable : 0
DNS Resolution Failures : 0
Reception Error         : 0
Transmission Error      : 0

IP-SLA Latest Probe Results
Last Probe Time        : 2018 Jul 13 02:00:35
Packets Sent           : 1
Packets Received       : 1
Packet Loss in Test    : 0.0000%

Minimum RTT(ms)        : 12
Maximum RTT(ms)        : 12
Average RTT(ms)        : 12
DNS RTT(ms)            : 0
TCP RTT(ms)            : 12

```

Showing history results for ip-sla:

```

switch# sh ip-sla abcd history-results
IP-SLA Name: abcd

Session Details
=====
IP-SLA Type           : tcp-connect      Status                : running

Probe Interval(seconds) : 30              History Interval(seconds) : 600

Source Port           : 3000              Destination Port        : 4000

Source IP Address      : 10.0.0.1
Dest Host Name/IP Address : 10.0.0.2

History Probe Results
=====
Packet Stats
-----
Probes Transmitted    : 4              Packets Sent           : 4
Packets Received      : 4              Loss Percentage        : 0.0000%

Error Stats
-----
Transmission Errors   : 0              Reception Errors       : 0
Bind Errors           : 0              Dest. Unreachable      : 0
Probes Timed-Out      : 0              DNS Resolution Failures : 0

Probe RTT Stats
-----
Min RTT(ms)          : 0              Max RTT(ms)           : 0
Avg RTT(ms)          : 0

DNS RTT Stats

```

```

-----
Min  RTT (ms)           :           Max  RTT (ms)           :
Avg  RTT (ms)           :

```

Command History

Release	Modification
10.16.1000	Added new parameter history-results . Updated to display history interval .
10.12.1000	Updated to display https as an IP-SLA type.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
8400	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

start-test

start-test

Description

Starts the IP-SLA probes.

Examples

```

switch(config)# ip-sla test
switch(config-ip-sla-test)# start-test

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	config-ip-sla-<IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

stop-test

stop-test

Description

Stops the IP-SLA probes.

Examples

```
switch(config)# ip-sla test
switch(config-ip-sla-test)# stop-test
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	config-ip-sla- <i><IP-SLA-NAME></i>	Administrators or local user group members with execution rights for this command.

tcp-connect

```
tcp-connect {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} <DEST-PORT-NUM> [history-
interval <HISTORY-INTERVAL>] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}
[probe-interval <PROBE-INTERVAL>] [source {<IPV4-ADDR>|<IPV6-ADDR>|<IFNAME>}}] [source-port
<PORT-NUM>]
```

```
no tcp-connect {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} <DEST-PORT-NUM> [history-
interval <HISTORY-INTERVAL>] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}
[probe-interval <PROBE-INTERVAL>] [source {<IPV4-ADDR>|<IPV6-ADDR>|<IFNAME>}}] [source-port
<PORT-NUM>]
```

Description

Configures TCP connect as the IP-SLA test mechanism. Requires destination address/hostname and destination port for the IP-SLA of tcp-connect IP-SLA type.

The **no** form of this command removes the the TCP connection.

Parameter	Description
{<DEST-IPV4-ADDR> <DEST-IPV6-ADDR> <HOSTNAME>}	Selects the destination, either an IPv4 address, an IPv6 address, or hostname, for the IP-SLA.
<DEST-PORT-NUM>	Destination port for the IP-SLA. Range: 1 to 65535.
history-interval <HISTORY-INTERVAL>	Configures the history interval for the IP-SLA. Set the history interval to minimum of two times the probe-interval for the SLA. Range: 60 to 7200.

Parameter	Description
name-server {<IPV4-ADDR-DNS-SERVER> <IPV6-ADDR-DNS-SERVER>}	Specifies the DNS server for destination hostname resolution.
probe-interval <PROBE-INTERVAL>	Probe interval in seconds. Range: 30 to 604800.
source {<IPV4-ADDR> <IPV6-ADDR> IFNAME>}	Selects the source, either an IPv4 address, an IPv6 address, or hostname for SLA probes.
source-port <PORT-NUM>	Specifies the port for the IP-SLA test.

Examples

Configuring TCP connect echo with parameters, including history interval:

```
switch(config)# ip-sla tcp
switch(config-ip-sla-tcp)# tcp-connect https://device.example.com 8080 name-server
10.10.10.2 history-interval 180 source 1/1/1 source-port 6000
```

Disabling the TCP connect:

```
switch(config-ip-sla-tcp)# no tcp-connect 10:1::1:1 8080 source 1/1/1 source-port
6000
```

Command History

Release	Modification
10.16.1000	Added new parameters history-interval . Also, IPv6 addresses can be used.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
8400	config-ip-sla-<IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

udp-echo

```
udp-echo {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} <DEST-PORT-NUM> [history-interval
<HISTORY-INTERVAL>] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}
[payload-size <PAYLOAD-SIZE>] [probe-interval <PROBE-INTERVAL>] [source {<IPV4-
ADDR>|<IPV6-ADDR>|IFNAME>}] [source-port <SOURCE-PORT-NUM>] [timeout <TIMEOUT>] [tos
<TYPE-OF-SERVICE>]
```

```
no udp-echo {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} <DEST-PORT-NUM> [history-
interval <HISTORY-INTERVAL>] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}]
```

```
[payload-size <PAYLOAD-SIZE>] [probe-interval <PROBE-INTERVAL>] [source {<IPV4-ADDR>|<IPV6-ADDR>|IFNAME}] [source-port <SOURCE-PORT-NUM>] [timeout <TIMEOUT>] [tos <TYPE-OF-SERVICE>]
```

Description

Configures UDP echo as the IP-SLA test mechanism. Requires destination address/hostname and destination port number for the IP-SLA of udp-echo SLA type.

The **no** form of this command removes the UDP echo configuration.

Parameter	Description
{<DEST-IPV4-ADDR> <DEST-IPV6-ADDR> <HOSTNAME>}	Selects the destination, either an IPv4 address, an IPv6 address, or hostname, for the IP-SLA.
<DEST-PORT-NUM>	Specifies the destination port for the IP-SLA. Range: 1 to 65535.
history-interval <HISTORY-INTERVAL>	Configures the history interval for the IP-SLA. Set the history interval to minimum of two times the probe-interval for the SLA. Range: 10 to 7200.
name-server {<IPV4-ADDR-DNS-SERVER> <IPV6-ADDR-DNS-SERVER>}	Specifies the DNS server for destination hostname resolution.
payload-size <PAYLOAD-SIZE>	Specifies the payload size of an SLA probe. Range: 28 to 1440.
probe-interval <PROBE-INTERVAL>	Specifies the probe interval in seconds. Range: 5 to 604800.
source {<IPV4-ADDR> <IPV6-ADDR> IFNAME}	Selects the source, either an IPv4 address, an IPv6 address, or hostname for SLA probes.
source-port <SOURCE-PORT-NUM>	Configures the source port for the IP-SLA test. Range: 1 to 65535.
timeout <TIMEOUT>	Specifies the interval before a probe is timed out. Range: 5 to 604800.
tos <TYPE-OF-SERVICE>	Specifies the type of service. Range: 0 to 255.

Examples

Configuring UDP echo with parameters, including history interval and timeout:

```
switch(config-ipsla-1)# udp-echo https://device.example.com 4000 history-interval 180 name-server 2.2.2.2 payload-size 100 probe-interval 90 source 4.4.4.4 source-
```

```
port 8000 timeout 20
```

Removing UDP echo configuration:

```
switch(config-ipsla-1)# no udp-echo https://device.example.com 8080 history-  
interval 160 name-server 10.10.10.2 payload-size 50 source 2.2.2.1 timeout 20
```

Command History

Release	Modification
10.16.1000	Added two new parameters history-interval and timeout . Also, IPv6 addresses can be used.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
8400	config-ip-sla-<IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

udp-jitter-voip

```
udp-jitter-voip {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} <DEST-PORT-NUM>  
[advantage-factor <ADVANTAGE-FACTOR>] [codec-type <CODEC-TYPE>] [history-interval  
<HISTORY-INTERVAL>] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}] [probe-  
interval <PROBE-INTERVAL>] [source {<IPV4-ADDR>|<IPV6-ADDR>|IFNAME}] [source-port  
<SOURCE-PORT-NUM>] [tos <TYPE-OF-SERVICE>]
```

```
no udp-jitter-voip {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} <DEST-PORT-NUM>  
[advantage-factor <ADVANTAGE-FACTOR>] [codec-type <CODEC-TYPE>] [history-interval  
<HISTORY-INTERVAL>] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}] [probe-  
interval <PROBE-INTERVAL>] [source {<IPV4-ADDR>|<IPV6-ADDR>|IFNAME}] [source-port  
<SOURCE-PORT-NUM>] [tos <TYPE-OF-SERVICE>]
```

Description

Configure UDP jitter VoIP as the IP-SLA test mechanism. Requires destination address/hostname and source address/interface for the IP-SLA of udp-jitter-voip IP-SLA type.

The **no** form of this command removes the UDP jitter VoIP configuration.

Parameter	Description
{<DEST-IPV4-ADDR> <DEST-IPV6-ADDR> <HOSTNAME>}	Selects the destination, either an IPv4 address, an IPv6 address, or hostname, for the IP-SLA.
<DEST-PORT-NUM>	Selects the port number for the IP-SLA. Range: 1 to 65535.

Parameter	Description
<code>advantage-factor <ADVANTAGE-FACTOR></code>	Selects the value for the advantage factor. Default value is 0. Range: 0 to 20.
<code>codec-type <CODEC-TYPE></code>	Selects the codec-type for the Voip IP-SLA test.
<code>history-interval <HISTORY-INTERVAL></code>	Configures the history interval for the IP-SLA. Set the history interval to minimum of two times the probe-interval for the SLA. Range: 240 to 7200.
<code>name-server {<IPV4-ADDR-DNS-SERVER> <IPV6-ADDR-DNS-SERVER>}</code>	Specifies the DNS server for destination hostname resolution.
<code>probe-interval <PROBE-INTERVAL></code>	Specifies the probe interval in seconds. Range: 120 to 604800.
<code>source {<IPV4-ADDR> <IPV6-ADDR> IFNAME}</code>	Selects the source, either an IPv4 address, an IPv6 address, or hostname for SLA probes.
<code>source-port <SOURCE-PORT-NUM></code>	Specifies the value of source port for the IP-SLA probes.
<code>tos <TYPE-OF-SERVICE></code>	Specifies the type of service. Range: 0 to 255.

Examples

Configuring udp-jitter-voip with optional parameters, including history interval:

```
switch(config-ipsla-1)# udp-jitter-voip https://device.arubanetworks.com 8080
advantage-factor 10 codec-type g711a history-interval 240 name-server 10.10.10.2
probe-interval 120 source 10.1.1.1 source-port 8888 tos 10
```

Configuring udp-jitter-voip with optional parameters:

```
switch(config-ipsla-1)# udp-jitter-voip 2.2.2.2 8080 advantage-factor 10 codec-
type g711a
```

Disabling udp-jitter-voip:

```
switch(config-ipsla-1)# no udp-jitter-voip https://device.example.com 8080
advantage-factor 10 codec-type g711a name-server 10.10.10.2 probe-interval 120
source 10.1.1.1 source-port 8888 tos 10
```

Command History

Release	Modification
10.16.1000	Added new parameters history-interval . Also, IPv6 addresses can be used.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
8400	config-ip-sla-<IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

vrf

```
vrf <VRF-NAME>
no vrf [<VRF-NAME>]
```

Description

Configures the VRF on which the SLA will send or receive packets. By default, the default VRF is used. The **no** form of the command removes VRF from SLA.

Parameter	Description
<VRF-NAME>	Specifies a VRF name. Length: Default: default. If no VRF name is specified, then the default VRF name will be taken.

Examples

```
switch(config-ip-sla-test)# vrf ipslasrc
```

```
switch(config-ip-sla-test)# no vrf
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	config-ip-sla-<IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

show interface

```
show interface [<IFNNAME>|<IFRANGE>] [brief | physical]
show interface [<IFNNAME>|<IFRANGE>] [extended [non-zero] | [human-readable]]
```



```
show interface [<IFNAME>] monitor [human-readable]
show interface [lag | loopback | tunnel | vlan ] [<ID>] [brief]
show interface lag [<LAG-ID>] [extended [non-zero] | [human-readable]]
show interface lag [<LAG-ID>] monitor [human-readable]
```

Description

Shows active configurations and operational status information for interfaces.

Parameter	Description
<IFNAME>	Specifies a interface name.
<IFRANGE>	Specifies the port identifier range.
brief	Shows brief info in tabular format.
physical	Shows the physical connection info in tabular format.
extended	Shows additional statistics, including the tx filtered and rx filtered counters. - Rx filter packets are protocol packets received when the protocol is disabled on the switch and there is only one port in the VLAN. Protocols include OSPF, PIM, RIP, LACP, and LLDP. - An example of a Tx filtered packet would be a multicast packet being filtered from going out of the ingress port.
human-readable	Shows statistics rounded to the nearest power of 1000, for example, 1K, 345M, 2G. This is available only in the CLI interface output.
non-zero	Shows only non zero statistics.
LAG	Shows LAG interface information.
monitor	Continuously monitor interface statistics.
LOOPBACK	Shows loopback interface information.
TUNNEL	Shows tunnel interface information.
VLAN	Shows VLAN interface information.
<LAG-ID>	Specifies the LAG number. Range: 1-256
<LOOPBACK-ID>	Specifies the LOOPBACK number. Range: 0-255
<TUNNEL-ID>	Specifies the tunnel ID. Range: 1-255
<VLAN-ID>	Specifies the VLAN ID. Range: 1-4094
VXLAN	Shows the VXLAN interface information.
<VXLAN-ID>	Specifies the VXLAN interface identifier. Default: 1

Examples

Showing interface information when it is configured as a route-only port:

```

switch# show interface 1/1/1
Interface 1/1/1 is up
Admin state is up
Link state: up for 2 days (since Sun Jun 21 05:30:22 UTC 2020)
Link transitions: 1
Description: backup data center link
Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
MTU 1500
Type 1GbT
Full-duplex
qos trust none
Speed 1000 Mb/s
Auto-negotiation is on
Flow-control: off
Error-control: off
L3 Counters: Rx Enabled, Tx Enabled
Rate collection interval: 300 seconds

```

Rates	RX	TX	Total (RX+TX)
Mbits / sec	0.00	0.00	0.00
KPkts / sec	0.00	0.00	0.00
Unicast	0.00	0.00	0.00
Multicast	0.00	0.00	0.00
Broadcast	0.00	0.00	0.00
Utilization %	0.00	0.00	0.00

Statistics	RX	TX	Total
Packets	0	0	0
Unicast	0	0	0
Multicast	0	0	0
Broadcast	0	0	0
Bytes	0	0	0
Jumbos	0	0	0
Dropped	0	0	0
Filtered	0	0	0
Pause Frames	0	0	0
L3 Packets	0	0	0
L3 Bytes	0	0	0
Errors	0	0	0
CRC/FCS	0	n/a	0
Collision	n/a	0	0
Runts	0	n/a	0
Giants	0	n/a	0
Other	0	0	0

Showing information when the interface is currently linked at a downshifted speed:

```

switch(config-if)# show interface 1/1/1
Interface 1/1/1 is up
...
Auto-negotiation is on with downshift active

```

Showing information when the interface is shut down during a VSX split:

```

switch(config-if)# show interface 1/1/1
Interface 1/1/1 is down
Admin state is up
State information: Disabled by VSX

```

```

Link state: down for 3 days (since Tue Mar 16 05:20:47 UTC 2021)
Link transitions: 0
Description:
Hardware: Ethernet, MAC Address: 04:09:73:62:90:e7
MTU 1500
Type SFP+DAC3
Full-duplex
qos trust none
Speed 0 Mb/s
Auto-negotiation is off
Flow-control: off
Error-control: off
VLAN Mode: native-untagged
Native VLAN: 1
Allowed VLAN List: 1502-1505
Rate collection interval: 300 seconds

```

Rate	RX	TX	Total (RX+TX)
Mbits / sec	0.00	0.00	0.00
KPkts / sec	0.00	0.00	0.00
Unicast	0.00	0.00	0.00
Multicast	0.00	0.00	0.00
Broadcast	0.00	0.00	0.00
Utilization	0.00	0.00	0.00

Statistic	RX	TX	Total
Packets	0	0	0
Unicast	0	0	0
Multicast	0	0	0
Broadcast	0	0	0
Bytes	0	0	0
Jumbos	0	0	0
Dropped	0	0	0
Pause Frames	0	0	0
Errors	0	0	0
CRC/FCS	0	n/a	0
Collision	n/a	0	0
Runts	0	n/a	0
Giants	0	n/a	0

Showing the monitor information:



In monitor mode, the CLI refreshes data automatically until it is exited by entering **q**. Pressing **?** opens the help menu to display which options are available in this context.

Interface 1/1/1 is up			
Rate	RX	TX	Total (RX+TX)
MBits / sec	30196.43	30196.43	60392.85
MPkts / sec	58977.39	58977.40	117954.79
Unicast	0.00	0.00	0.00
Multicast	58977.39	58977.40	117954.79
Broadcast	0.00	0.00	0.00
Utilization %	75.49	75.49	150.98
Statistic	RX	TX	Total (RX+TX)
Packets	4756527649	4756527865	9513055514

```

Unicast                0                0                0
Multicast              4756527649          4756527865          9513055514
Broadcast              2                0                2
Bytes                 304417778668          304417795428          608835574096
Jumbos                0                0                0
Dropped               0              19028847730          19028847730
Pause Frames          0                0                0
Errors                0                0                0
CRC/FCS               0                n/a                0
help: ?, quit: q

```

```

Help for Interface Monitor
h  Toggle human-readable mode
c  Clear interface statistics
Does not apply to rates
Arrows, PgUp, PgDn, Home, End
Navigate interface statistics
Delay: 2
help: ?, quit: q

```

Showing the output for interface 1/1/1 in human-readable format:



In human-readable format, the **< 1** symbol for **Utilization** indicates that the amount of packets is between zero and one. This is true in cases where the number of bytes increases but the number of packets and the **Utilization** value is not displayed even in the normal output, where the human-readable parameter is not included in the command.

```

switch(config-if)# show interface 1/1/1 human-readable
Interface 1/1/1 is up

```

Rate	RX	TX	Total (RX+TX)
Bits / sec	3M	3M	6M
Pkts / sec	316	316	633
Unicast	319	319	638
Multicast	0	0	0
Broadcast	0	0	0
Utilization %	< 1	< 1	< 1
Statistic	RX	TX	Total
Packets	577K	577K	1M
Unicast	577K	577K	1M
Multicast	0	51	51
Broadcast	0	15	15
Bytes	744M	745M	1G
Jumbos	0	0	0
Dropped	0	0	0
Filtered	0	0	0
Pause Frames	0	0	0
Errors	0	0	0
CRC/FCS	0	n/a	0
Collision	n/a	0	0
Runts	0	n/a	0
Giants	0	n/a	0

Showing information about extended counters:



The output of the `show interface extended` command varies depending on the switch model and configuration.

```
switch(config-if)# show interface 1/1/17 extended
-----
Interface 1/1/17
-----
Statistics                                     Value
-----
Dot1d Tp Port In Frames                      547
Dot1d Tp Port Out Frames                     608
Dot3 In Pause Frames                         0
Dot3 Out Pause Frames                       0
Ethernet Stats Broadcast Packets             19
Ethernet Stats Bytes                        40162
Ethernet Stats Packets                      342
...
-----
Error-Statistics                             Value
-----
Dot1d Base Port MTU Exceeded Discards        0
Dot3 Control In Unknown Opcodes              0
Dot3 Stats Alignment Errors                  0
Dot3 Stats FCS Errors                       0
Dot3 Stats Frame Too Longs                  0
Dot3 Stats Internal Mac Transmit Errors      0
Ethernet RX Oversize Packets                 0
...
```

Showing interface link-status:

```
switch# show interface link-status
-----

Port          Type          Physical Link      Last
Link State   Transitions Change
-----
1/1/1         1G-BT         down      0         --
1/1/2         1G-BT         up        1         1 minute ago (Fri Mar 09
12:36:56 UTC 2018)
1/1/3         1G-BT         up        1         1 minute ago (Fri Mar 09
12:36:56 UTC 2018)
1/1/4         --            down      0         --
1/1/5         --            down      0         --
```

Showing interface loopback 1 link-status:

```
-----

Port          Type          Physical Link      Last
Link State   Transitions Change
-----
loopback1     --            up        --        --
```

Showing interface 1/1/2-1/1/3 link-status:

Port	Type	Physical Link State	Link Transitions	Last Change
1/1/2	1G-BT	up	1	1 minute ago (Fri Mar 09 12:36:56 UTC 2018)
1/1/3	1G-BT	up	1	1 minute ago (Fri Mar 09 12:36:56 UTC 2018)

Showing interface link-status:

```
switch# show interface link-status
```

Port	Type	Physical Link State	Link Transitions	Link Flaps Ignored	Last Change
1/1/1	1G-BT	down	0	0	--
1/1/2	1G-BT	up	1	0	1 minute ago (Fri Mar 09 12:36:56 UTC 2018)
1/1/3	1G-BT	up	1	0	1 minute ago (Fri Mar 09 12:36:56 UTC 2018)
1/1/4	--	down	0	0	--
1/1/5	--	down	0	0	--

Showing state information when interface is blocked:

```
8360(config-if)# show interface 1/1/1
```

Interface 1/1/1 is up (Blocked)
Admin state is up
State information: Blocked by UDLD
Link state: up for 1 minute (since Mon Jun 10 09:25:27 UTC 2024)
Link transitions: 1
Description:
Persona:
Hardware: Ethernet, MAC Address: 00:fd:45:67:85:91
MTU 1500
Type 10G-LR / 10G SFP+ LR
Full-duplex
qos trust none
Speed 10000 Mb/s
Auto-negotiation is off
Flow-control: off
Error-control: off
VLAN Mode: access
Access VLAN: 1
Rate collection interval: 300 seconds

Rate	RX	TX	Total (RX+TX)
Mbits / sec	0.00	0.00	0.00
KPkts / sec	0.00	0.00	0.00
Unicast	0.00	0.00	0.00
Multicast	0.00	0.00	0.00
Broadcast	0.00	0.00	0.00
Utilization %	0.00	0.00	0.00
Statistic	RX	TX	Total
Packets	15	15	30

Unicast	12	12	24
Multicast	3	3	6
Broadcast	0	0	0
Bytes	1350	1350	2700
Jumbos	0	0	0
Dropped	0	0	0
Pause Frames	0	0	0
Errors	0	0	0
CRC/FCS	0	n/a	0
Collision	n/a	0	0
Runts	0	n/a	0
Giants	0	n/a	0

Command History

Release	Modification
10.15	Added state information when port goes into down state.
10.11	Added <code>monitor</code> parameter.
10.10	Added <code>human-readable</code> parameter.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

show interface statistics

```

show interface [<IFNAME>|<IFRANGE>] statistics [non-zero] [human-readable]
show interface [<IFNAME>|<IFRANGE>] statistics monitor [non-zero] [human-readable]
show interface [<IFNAME>|<IFRANGE>] error-statistics [non-zero] [human-readable]
show interface [<IFNAME>|<IFRANGE>] error-statistics monitor [non-zero] [human-readable]
show interface lag [<LAG-ID>] statistics [non-zero] [human-readable]
show interface lag [<LAG-ID>] statistics monitor [non-zero] [human-readable]
show interface lag [<LAG-ID>] error-statistics [non-zero] [human-readable]
show interface lag [<LAG-ID>] error-statistics monitor [non-zero] [human-readable]
show interface vxlan <VXLAN-ID> statistics [non-zero] [human-readable]

```

Description

Shows statistics for switch interfaces such as packets transmitted and received, bytes transmitted and received, broadcast and multicast packets.

Parameter	Description
<code><IFNAME></code>	Specifies a interface name.
<code><IFRANGE></code>	Specifies the port identifier range.

Parameter	Description
LAG	Shows LAG interface information.
<LAG-ID>	Specifies the LAG number. Range: 1-256
VXLAN	Shows the VXLAN interface information.
<VXLAN-ID>	Specifies the VXLAN interface identifier. Default: 1
monitor	Continuously monitor interface statistics.
human-readable	Shows statistics rounded to the nearest power of 1000, for example, 1K, 345M, 2G.
non-zero	Shows only non zero statistics.

Examples

Showing statistics of all interfaces:

```
show interface statistics
```

Interface	RX Bytes	RX Packets	RX Drops	TX Bytes	TX Packets	TX Drops	RX Broadcast	RX Multicast	TX Broadcast	TX Multicast	RX Pause	TX Pause
1/1/1	2727136	1975	0	17796	195	0	82	1788	96	54	0	0
1/1/10	0	0	0	0	0	0	0	0	0	0	0	0
1/1/11	0	0	0	0	0	0	0	0	0	0	0	0
1/1/12	0	0	0	0	0	0	0	0	0	0	0	0
...												
1/1/30 - lag1	0	0	0	11271	92	0	0	0	0	51	0	0
1/1/31 - lag2	2360	25	50	2732119	2040	0	0	0	178	1839	0	0
1/1/32 - lag2	0	0	0	11373	93	0	0	0	0	51	0	0
vlan1	0	0	0	0	0	0	0	0	0	0	0	0

Showing statistics of all interfaces with only non-zero statistics:

```
show interface statistics non-zero
```

Interface	RX Bytes	RX Packets	RX Drops	TX Bytes	TX Packets	TX Drops	RX Broadcast	RX Multicast	TX Broadcast	TX Multicast	RX Pause	TX Pause
1/1/1	2727136	1975	0	17796	195	0	82	1788	96	54	0	0
1/1/30 - lag1	0	0	0	11271	92	0	0	0	0	51	0	0
1/1/31 - lag2	2360	25	50	2732119	2040	0	0	0	178	1839	0	0
1/1/32 - lag2	0	0	0	11373	93	0	0	0	0	51	0	0

Showing statistics of all interfaces in the human-readable format:

```
show interface statistics human-readable
```

Interface	RX Bytes	RX Pkts	RX Drops	TX Bytes	TX Pkts	TX Drops	RX Bcast	RX Mcast	TX Bcast	TX Mcast	RX Pause	TX Pause
1/1/1	744M	577K	0	745M	578K	0	0	0	73	287	0	0
1/1/2	474M	367K	0	475M	369K	0	0	0	73	288	0	0
1/1/3	0	0	0	0	0	0	0	0	0	0	0	0

Showing statistics of a single interfaces:

```
show interface 1/1/2 statistics
```

Interface	RX Bytes	RX Packets	RX Drops	TX Bytes	TX Packets	TX Drops	RX Broadcast	RX Multicast	TX Broadcast	TX Multicast	RX Pause	TX Pause
1/1/2	2725080	1931	0	25877	253	0	21	1788	65	55	0	0

Showing statistics of all members of a LAG interface:

```
show interface lag1 statistics
```

Interface	RX Bytes	RX Packets	RX Drops	TX Bytes	TX Packets	TX Drops	RX Broadcast	RX Multicast	TX Broadcast	TX Multicast	RX Pause	TX Pause
1/1/3 - lag1	2424	26	0	2734082	2062	0	0	0	191	1848	0	0
1/1/30 - lag1	0	0	0	12383	100	0	0	0	0	59	0	0
lag1	2424	26	0	2746465	2162	0	0	0	191	1907	0	0

Showing error statistics of all interfaces:

```
show interface error-statistics
```

Interface	RX Errors	TX Errors	Giants	Runts	CRC/FCS	Collisions
1/1/1	190	20	100647	0	0	0
1/1/10	0	0	100	290	7165	949
1/1/11	0	0	0	0	0	0
1/1/12	0	0	0	0	0	0
...						
1/1/30 - lag1	1500	500	45800	0	0	0
1/1/31 - lag2	0	0	11	27	0	0
1/1/32 - lag2	0	0	0	0	6	18

Showing monitor statistics:



The rows and columns of show interface monitor statistics depends on the length of width of the client terminal. The CLI can be navigated using the arrow keys as well as the PageUp, PageDown, Home, and End keys.

```
show interface statistics monitor
```

Interface	RX Bytes	RX Packets >>
1/1/1	3440525421984	53758209526
1/1/2	3440526607008	53758228042
1/1/3	3440527785312	53758246453
1/1/30	3440559671264	53758744653
1/1/31	3440560851680	53758763098
1/1/32	3440562028704	53758781489

```
help: ?, quit: q
```

Help for Interface Monitor

```
f  Toggle full statistics
h  Toggle human-readable mode
n  Toggle non-zero mode
r  Toggle rate display
```

```
c  Clear interface statistics
    Does not apply to rates
```

```
Arrows, PgUp, PgDn, Home, End
    Navigate interface statistics
```

```
Delay:2
```

```
help: ?, qui
```

Showing monitor error statistics in human-readable format:

```

show interface 1/1/1-1/1/3,1/1/30-1/1/32 error-statistics monitor human-readable
-----
Interface           RX Errors   TX Errors   RX Giants   RX Runts   CRC/FCS   Collisions
-----
1/1/1                0           0           0           0           0           0
1/1/2                0           0           0           0           0           0
1/1/3                0           0           0           0           0           0
1/1/30               0           0           0           0           0           0
1/1/31               0           0           0           0           0           0
1/1/32               0           0           0           0           0           0

Human-readable                                           help: ?, quit: q

Help for Interface Monitor

h    Toggle human-readable mode
n    Toggle non-zero mode

c    Clear interface statistics
      Does not apply to rates

Arrows, PgUp, PgDn, Home, End
      Navigate interface statistics

Delay:2
                                           help: ?, quit: q

```

Command History

Release	Modification
10.11	Added <code>monitor</code> parameter.
10.10	Added <code>human-readable</code> parameter.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

Mirroring allows you to replicate all traffic arriving and/or leaving the selected system interfaces. This data can be used for collection or analysis.

The traffic replicated using mirroring can be sent to a separate interface on the same switch as the traffic source for analysis or inspection. Such a collection of interfaces and settings is called a mirror session.

A mirror session can be configured with many traffic sources but only a single output, or destination. In the initial configuration, the mirror session is disabled. You have enable the feature to start the replication.



Care must be taken in choosing the number and rates of sources to avoid over-saturating a session destination. A mirror session with multiple 10G sources can overwhelm a single 10G destination and important data may be lost.

Mirroring and sFlow

The mirroring feature (when mirroring received traffic) and the sFlow sampling feature both require the receive (rx) capability of a given port. If both features are configured and enabled to use the receive capability on the same port, only one of the features can perform its task.

This interaction does not affect transmit (tx) mirroring because sFlow does not use the transmit (tx) capability of a port.

Behavior if sFlow is enabled and mirror enable is attempted

If sFlow is enabled on a port and a mirroring session specifies the same port as a source of received traffic (the source is configured with a direction of `rx` or `both`):

- The attempt to enable the mirroring session fails and an error is returned.
- To enable the mirroring session, first you must disable sFlow on that port.

Behavior if mirroring is enabled and sFlow enable is attempted

If a mirroring session specifies a port as a source of received traffic (the source is configured with a direction of `rx` or `both`), and you attempt to enable sFlow on the same port:

- Mirroring on that port continues.
- No error or warning message is returned when sFlow is enabled, but sFlow sampling on that port does not occur.

When sFlow is enabled on a port but sampling is not occurring, the `show sflow <INTERFACE-NAME>` command shows that sFlow is enabled but shows a value of 0 (zero) for the number of samples.

To activate sFlow sampling on that port, you must do the following:

1. Disable the mirroring session on the port.
2. Disable sFlow on the port.
3. Enable sFlow on the port.

Behavior when the startup configuration has both sFlow and rx mirroring enabled on the same port

If the startup configuration has the same port configured with both sFlow enabled and as a source of received traffic in an enabled mirroring session:

- During a boot or management module failover operation, it is not possible to predict whether the receive capability of the port will be assigned to the sFlow feature or to the mirroring feature.
- To ensure that the feature that you want is used on a specific port, after the boot operation or management module failover operation completes, disable both features on that port and then enable the feature you want to use.

Mirror statistics

Mirror statistics are reset for a Mirror-to-CPU session when an interface is added or removed from a LAG that is a source interface in the Mirror session and during a failover.

Mirror statistics are reset for a Mirror-to-CPU session on a failover.

Classifier policies and mirroring sessions

Network traffic can be mirrored to a destination interface in two ways:

- Using a mirroring session alone.
- Using Classifier Policies with mirror actions in conjunction with a mirroring session.

Basic mirroring sessions provide coarse control over the type of traffic mirrored from a source: all received, all transmitted, or both. However, a traffic class within a Classifier Policy applied to a source can provide much finer grained control of mirrored traffic. For example, a policy can match on many different aspects of the Ethernet or IPv4 or IPv6 header information in each frame or packet received or transmitted on an interface.

The steps to configure a policy and class with a mirror action are the following:

1. Configuring a mirroring session with a destination interface.
2. Enabling the mirroring session.
3. Configuring the Classifier Policy, specifying the mirroring session ID in the mirror action.

Any subsequent configuration changes to either the enabled mirroring session or the classifier policy can affect the behavior of the network monitoring that occurs. For examples, see Scenario 1 and Scenario 2.

Scenario 1

1. Mirroring session 1 is configured with destination interface 1/1/10 and source interface 1/1/5 in the `both` direction, then the session is enabled.
2. Mirroring session 2 is configured with destination interface 1/1/20, then the session is enabled.
3. Policy `Policy_2` is configured with a class matching OSPF traffic from any source IPv4 address to

any destination IPv4 address and an action of `mirror`, specifying mirroring session 2.

4. `Policy_2` is applied to interface 1/1/5 in the inbound direction.

This sequence of actions creates a situation where the interface 1/1/5 is effectively configured as a source for two separate enabled mirroring sessions. This configuration is not permitted if you attempt to configure and enable the two mirroring sessions through the CLI. However, mirroring may occur for both sessions because policies with mirror actions have priority over basic mirroring sessions.

In this example:

- Because of `Policy_2`, all OSPF traffic ingressing interface 1/1/5 is mirrored to 1/1/20, which is the destination interface of mirroring session 2.
- After `Policy_2` is applied, and because of the mirroring session 1 is enabled, all non-OSPF traffic ingressing interface 1/1/5 is mirrored to 1/1/10, which is the destination interface of mirroring session 1.
- Because `Policy_2` does not match egressing traffic, and because mirroring session 1 is enabled, all traffic egressing interface 1/1/5 is mirrored to 1/1/10, which is the destination interface of mirroring session 1.

Scenario 2

1. Mirroring session 2 is configured with destination interface 1/1/20 and source interface 1/1/3, then the session is enabled.
2. Policy `Policy_2` is configured with a class matching OSPF traffic from any source IPv4 address to any destination IPv4 address and an action of `mirror` specifying mirroring session 2.
3. `Policy_2` is applied to interface 1/1/5 in the inbound direction.

In this scenario, a single mirroring session is configured with a source interface and is configured as the target of the mirror action of a policy applied to a different source interface. In this example, the destination interface 1/1/20 receives traffic from interface 1/1/3 and from interface 1/1/5.

Mirroring commands

clear mirror

```
clear mirror [all | <SESSION-ID>]
```

Description

Clears the mirror statistics for all configured mirror sessions or a specified session

Parameter	Description
<code>all</code>	Specifies all configured sessions.
<code><SESSION-ID></code>	Specifies a numeric identifier for the session. Range: 1 to 4

Examples

Clearing mirror statistics for all configured mirror sessions:

```
switch# clear mirror all
```

Clearing mirror statistics for mirror session 1:

```
switch# clear mirror 1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

comment

```
comment <COMMENT>  
no comment
```

Description

Specifies a comment for the mirroring session.

The **no** form of this command removes the comment.

Parameter	Description
<COMMENT>	A comment string of up to 64 characters composed of letters, numbers, underscores, dashes, spaces, and periods.

Usage

Comments are optional and can be added or removed at any time without affecting the state of the mirroring session.

Adding a comment to a session that already has a comment replaces the existing comment.

Examples

Adding a comment to a mirror session:

```
switch(config-mirror-3)# comment This Mirror will be removed during next  
maintenance window
```

Removing the comment from mirror session 3:

```
switch(config-mirror-3)# no comment
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-mirror-<SESSION-ID>	Administrators or local user group members with execution rights for this command.

copy tcpdump-pcap

copy tcpdump-pcap <FILE-NAME> <REMOTE-URL>

Description

Saves packet capture files to external storage.

Parameter	Description
<FILE-NAME>	Specifies the packet capture file to save.
<REMOTE-URL>	Specifies the external storage to which the packet capture file will be saved.

Usage

Only four files can be saved at any point on the switch. Packet capture files are not saved after a failover or reboot. View a list of saved files using **diag utilities list-files**.

Examples

Saving my_capture_file.pcap to sftp://root@10.0.0.2/file.pcap:

```
switch# copy tcpdump-pcap my_capture_file.pcap sftp://root@10.0.0.2/file.pcap
root@10.0.0.2's password:
Connected to 10.0.0.2.
sftp > put my_capture_file.pcap file.pcap
Uploading my_capture_file.pcap to /root/file.pcap
my_capture_file.pcap          100%   156   219.8KB/s   00:00
Copied successfully.
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
8400	Manager (#)	Administrators or local user group members with execution rights for this command.

copy tshark-pcap

copy tshark-pcap <REMOTE-URL> [vrf <VRF-NAME>]

Description

Copies the tshark capture data to a file on a TFTP or SFTP server.

Parameter	Description
<REMOTE-URL>	Specifies the capture file on a remote TFTP or SFTP server. The URL syntax is: {tftp:// sftp://<USER>@} {<IP> <HOST>} [:<PORT>] [:blocksize=<SIZE>]/<FILE>
vrf <VRF-NAME>	Specifies the name of a VRF. Default: default.

Example

Copying the capture data to a file on SFTP server 10.0.0.2:

```
switch# copy tshark-pcap sftp://root@10.0.0.2/file.pcap

root@10.0.0.2's password:
Connected to 10.0.0.2.
sftp> put packets.pcap file.pcap
Uploading packets.pcap to /root/file.pcap
packets.pcap                               100% 156   219.8KB/s   00:00
Copied successfully.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	Manager (#)	Administrators or local user group members with execution rights for this command.

destination cpu

destination cpu
no destination cpu

Description

The command causes the mirror session to transmit mirrored packets to the switch CPU. This destination may be configured for multiple sessions, however only one such configured session may be active at a given time.

The diagnostic utility Tshark may be used to view and capture packets transmitted to the CPU through this route. Ctrl+C must be entered to terminate a Tshark capture session. More details can be found in the **Supportability Guide**.

The **no** form of this command will immediately stops mirroring traffic to the CPU, but will not remove any sources from the mirror configuration.

Examples

Configuring a mirror session with CPU as the destination.

```
switch# config
switch(config)# mirror session 1
switch(config-mirror-1)# destination cpu
```

Removing the destination entirely.

```
switch(config-mirror-1)# no destination cpu
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	config-mirror-<SESSION-ID>	Administrators or local user group members with execution rights for this command.

destination interface

```
destination interface {<INTERFACE-ID>|<LAG-NAME>}
no destination interface {<INTERFACE-ID>|<LAG-NAME>}
```

Description

Configures the specified interface as the destination of the mirrored traffic.

The **no** form of this command immediately disables the mirroring session and removes the specified destination interface from the configuration.

Parameter	Description
<INTERFACE-ID>	Specifies a interface. Format: member/slot/port.
<LAG-NAME>	Specifies a LAG (link aggregation group) identifier.

Usage

Supported mirror destinations: Layer 2 or Layer 3 Ethernet ports, LAGs, tunnel, or CPU as a Mirror Destination. A port that is already a member of a LAG is not a valid mirror destination.

Configuring a different destination interface in an enabled mirroring session causes all mirrored traffic to use the new destination interface. This action might cause a temporary suspension of mirrored source traffic during the reconfiguration.

Examples

Configuring a mirroring session and adding an interface as a destination:

```
switch(config)# mirror session 1
switch(config-mirror-1)# destination interface 1/1/1
```

Replacing the existing destination with different interface:

```
switch(config-mirror-1)# destination interface 1/1/12
```

Removing a destination:

```
switch(config-mirror-1)# no destination interface 1/1/12
```

Switch	Destination interface limit per mirror session (4 possible sessions)
8400	1

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-mirror- <i><SESSION-ID></i>	Administrators or local user group members with execution rights for this command.

destination tunnel

```
destination tunnel <TUNNEL-IPV4-ADDR> source <SOURCE-IPV4-ADDR>
  dscp <DSCP-VALUE> vrf <VRF-NAME> id <SPAN-ID>
no destination tunnel
```

Description

Specifies the tunnel where all mirrored traffic for the session is transmitted. Only one tunnel destination is allowed per session.

You may configure multiple mirror sessions with the same source/destination IP address pair, however, only one of those sessions sharing the same source/destination IP address pair can be enabled at a given time.

Multiple Mirror Sessions can be enabled with the same source/destination IP address pair if the span IDs are different for sessions. By default it is assigned 0 if not specified.

ERSPAN is not supported leaving the switch by the OOB port. If VRF management is configured for an ERSPAN session, the session will be in "mirror_err_tunnel_oob_port_not_supported" operation status.

ERSPAN is not supported leaving the switch encapsulated within another tunnel (e.g. GRE IPv4). When the path to the destination IP address will leave via a tunnel, the session will be in "tunnel_route_resolution_not_populated" operation status.



The interface/LAG used to transmit ERSPAN packets should not be a source in the same mirror session.

The **no** form of this command will cease the use of the tunnel and disable the session.

Parameter	Description
<code><TUNNEL-IPv4-ADDR></code>	Specifies the tunnel address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.
<code><SOURCE-IPv4-ADDR></code>	Specifies the source address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.
<code><DSCP-VALUE></code>	Specifies the DSCP value to be carried within the DS field of ERSPAN packet header. Range: 0 to 63. Default: 0.
<code><VRF-NAME></code>	Specifies a VRF name. Default: default.
<code><SPAN-ID></code>	Specifies the span ID for the ERSPAN session and during a failover. Range: 0 to 10.

Examples

Creating a Mirror Session and adding tunnel destination, source, dscp, and VRF:

```
switch# config
switch(config)# mirror session 1
switch(config-mirror-1)# destination tunnel 1.1.1.1 source 2.2.2.2 dscp 10 vrf
default
```

Replacing the existing tunnel destination:

```
switch(config-mirror-1)# destination tunnel 11.12.13.14 source 2.2.2.2 dscp 10 vrf
default
```

Replacing the existing destination with a different DSCP value:

```
switch(config-mirror-1)# destination tunnel 11.12.13.14 source 2.2.2.2 dscp 2 vrf
default
```

Removing the destination:

```
switch(config-mirror-1)# no destination tunnel
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	config-mirror-<SESSION-ID>	Administrators or local user group members with execution rights for this command.

diagnostic

diagnostic

```
diag utilities tshark [file]
diag utilities tshark [delete-file]
```

Description

Captures packets from a mirror-to-cpu session, and save the most recent 32MB to pcap file which can then be copied and analyzed. When capturing a mirror-to-cpu session to a file, packets will not be dumped to the console.



The `diagnostic` command must be entered prior to the `diag utilities tshark` command.

Use the **delete-file** form of this command to delete the most recent capture file.

Since **file** and **delete-file** are optional, the behavior of the base command **diag utilities tshark** does **not** save anything to a file, and instead dumps the tshark session to the console until **CTRL + c** is entered.

Parameter	Description
file	Saves captured packets to a temporary file.
delete-file	Deletes the most recent captured file.

Example

Performing diagnostic:

```
switch# diagnostic

switch# diagnostic utilities tshark file
Inspecting traffic mirrored to the CPU until Ctrl-C is entered
^CEnding traffic inspection.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

diag utilities tcpdump

```
diag utilities tcpdump [command <TEXT> | delete file <FILE-NAME> | list-files |
  vrf <VRF-NAME> | count <COUNT-NUM> | proto <PROTO-NUM> | host-ip <IP-ADDR> | source-ip
  <IP-ADDR> | destination-ip <IP-ADDR> | host-port <PORT> | source-port <PORT> |
  destination-port <PORT> | verbosity <LEVEL> | print <DATA> | ethernet-type <ETH-NUM>]
```

Description

Captures traffic received or transmitted over a network.

Parameter	Description
command <TEXT>	Captures packets based on a specified tcpdump command string.
delete file <FILE-NAME>	Deletes specified tcpdump list files.
list-files	Lists all the tcpdump capture files saved on the device.
vrf <VRF-NAME>	Captures packets on the specified VRF. If no VRF is named, the default is used.
count <COUNT-NUM>	Runs the tcpdump command until the specified number of packets are captured. Range: 1-2147483647.
proto <PROTO-NUM>	Captures packets of a particular type based on IP protocol number. Range: 0-255.
host-ip <IP-ADDR>	Captures packets matching with the source or destination IP address.
source-ip <IP-ADDR>	Captures packets from the specified IP address.
destination-ip <IP-ADDR>	Captures packets sent to the specified IP address.
host-port <PORT>	Captures packets matching with the source or destination port.
source-port <PORT>	Captures packets from the specified IP port.
destination-port <PORT>	Captures packets sent to the specified IP port.
verbosity <LEVEL>	Captures packets of the specified verbosity. Range: level1-level4. If no verbosity is specified, the default is level1.
print <DATA>	Captures the data of each packet. The maximum is 262144 bytes
ethernet-type <ETH-NUM>	Captures packets based on the particular ethernet type. Range: 0-65535.

Usage

- When using the **command** option, the only traffic captured will be packets that have been mirrored to the CPU.

- When using the **command** option, command line sanitization is performed to prevent options that may cause harm or security issues. The following options are blocked:
 - -i/--interface
 - -Z
 - -B/--buffer-size
 - -C
 - -W
 - -Z/--relinquish privileges
- Non-word operators such as "&" or "|" are not allowed. Use boolean keywords such as "and," "or," and "not."
- When using **command -r** to read a file, do not provide any directory path characters. Use list-files command to get the list of file names currently saved on the device, and then use those file names.
- A total of four files can be saved at any given point on the device. Packet capture files are not saved after a failover or reboot, but can be saved to external storage using the **copy tcpdump-pcap** command.

Examples

Inspecting traffic mirrored to the CPU via tcpdump and saving the output to my_capture_file.pcap:

```
switch# diag utilities tcpdump command -c 2 -x -w my_capture_file.pcap
Inspecting traffic mirrored to the CPU via tcpdump until Ctrl-C is entered.
2 packets captured
2 packets received by filter
0 packets dropped by kernel
Ending traffic capture.
```

Listing saved capture files:

```
switch# diag utilities tcpdump list-files
my_capture_file.pcap
```

Reading my_capture_file.pcap:

```
switch# diag utilities tcpdump command -r my_capture_file.pcap
reading from file /tmp/tcpdump/my_capture_file1.pcap, link-type EN10MB (Ethernet)
 1 11:59:34.047867 IP6 localhost.40318 > localhost.ntp: NTPv2, Reserved, length
12
    0x0000: 0000 0304 0006 0000 0000 0000 0000 86dd .....
    0x0010: 608a 7e47 0014 1140 0000 0000 0000 0000 `..~G...@.....
    0x0020: 0000 0000 0000 0001 0000 0000 0000 0000 .....
    0x0030: 0000 0000 0000 0001 9d7e 007b 0014 0027 .....~.{... '
    0x0040: 1601 0001 0000 0000 0000 0000 .....
 2 11:59:34.047915 IP6 localhost.ntp > localhost.40318: NTPv2, Reserved, length
12
    0x0000: 0000 0304 0006 0000 0000 0000 0000 86dd .....
    0x0010: 6b8d 23c5 0014 1140 0000 0000 0000 0000 k.#....@.....
    0x0020: 0000 0000 0000 0001 0000 0000 0000 0000 .....
    0x0030: 0000 0000 0000 0001 007b 9d7e 0014 0027 .....{.~... '
    0x0040: d681 0001 c016 0000 0000 0000
```

Removing my_capture_file.pcap:

```
switch# diag utilities tcpdump delete-file my_capture_file.pcap
Successfully removed file
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
8400	Manager (#)	Administrators or local user group members with execution rights for this command.

disable

disable

Description

Disables the mirroring session specified by the current command context.

Usage

By default, mirroring sessions are disabled.

When a mirroring session is disabled, the **show mirror** command for that session ID shows an **Admin Status** of **disable** and an **Operation Status** of **disabled**.

Example

Disabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3)# disable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-mirror- <i><SESSION-ID></i>	Administrators or local user group members with execution rights for this command.

enable

enable

Description

Enables the mirroring session for the current command context.

Usage

By default, mirroring sessions are disabled.

When a mirroring session is enabled, the **show mirror** command for that session ID shows an **Admin Status** of **enable** and an **Operation Status** of **enabled**.

If sFlow is enabled on an interface and a mirroring session specifies the same interface as the source of received traffic (the source is configured with a direction of **rx** or **both**):

- The attempt to enable the mirroring session fails and an error is returned.
- To enable the mirroring session, first you must disable sFlow on the port.



When adding, removing, or changing the configuration of a source interface in an enabled mirroring session, packets from other mirror sources using the same destination interface might be interrupted.

Example

Configuring and enabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3) # source interface 1/1/2 rx
switch(config-mirror-3) # destination interface 1/1/3
switch(config-mirror-3) # comment Monitor router port ingress-only traffic
switch(config-mirror-3) # enable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-mirror-<SESSION-ID>	Administrators or local user group members with execution rights for this command.

mirror session

```
mirror session <SESSION-ID>
no mirror session <SESSION-ID>
```

Description

Creates a mirroring session configuration context or enters an existing mirroring session configuration context.

From this context, you can enter commands to configure and enable or disable the mirroring session.

The **no** form of this command removes an existing mirroring session from the configuration.

Parameter	Description
<SESSION-ID>	Specifies the session identifier. Range: 1 to 4

Examples

```
switch(config)# mirror session 1
switch(config-mirror-1)#

switch(config)# mirror session 3
switch(config-mirror-3)#

switch(config)# no mirror session 1
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show mirror

```
show mirror [<SESSION-ID>] [vsx-peer]
```

Description

Shows information about mirroring sessions. If <SESSION-ID> is not specified, then the command shows a summary of all configured mirroring sessions. If <SESSION-ID> is specified, then the command shows detailed information about the specified mirroring session.

Parameter	Description
<SESSION-ID>	Specifies the session identifier. Range: 1 to 4
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Usage

Information in the **Admin Status** column of the command output indicates the configured status. The admin status is one of the following values:

- **enable:** The mirroring session is enabled.
- **disable:** The mirroring session has been configured but not yet enabled, or has been disabled.

Information in the **Operation Status** column indicates the status of the mirroring session. Operation status is one of the following values:

- **dest_doesnt_exist:** The configured destination interface is not found in the system. The mirroring session cannot be enabled.
- **destination_shutdown:** **The mirroring session is enabled, but the destination interface is shut down. No traffic** can be monitored.
- **disabled:** The mirroring session is disabled and is not in an error condition.
- **enabled:** The mirroring session is enabled.
- **external/driver_error:** An internal ASIC hardware error occurred.
- **hit_active_sessions_capacity:** The mirroring session could not be enabled because the maximum number of supported mirroring sessions are already enabled.
- **internal_error:** An invalid parameter was passed to the ASIC software layer.
- **no_dest_configured:** The mirroring session does not have a destination interface configured.
- **no_name_configured:** A software error occurred. The mirroring session does not have a session ID in its configuration.
- **null_mirror:** A software error occurred. The session object reference is invalid.
- **out_of_memory:** The system is out of memory, reboot recommended.
- **tunnel_route_resolution_not_populated:** If the destination tunnel IP address is not reachable.
- **unknown_error:** An unexpected error occurred.

Examples

Showing summary information about all configured mirroring sessions:

```
switch# show mirror
ID  Admin Status  Operation Status
---  -
1   enable       enabled
2   disable      disabled
3   disable      disabled
4   enable       internal_error
```

Showing detailed information about a single mirroring session:

```
switch# show mirror 3
Mirror Session: 3
Admin Status: disable
Operation Status: disabled
Comment: Monitor router port ingress-only traffic
Source: interface 1/1/2 rx
Destination: interface 1/1/3
switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

source interface

```
source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]
no source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]
```

Description

Configures the specified interface (either an Ethernet port or a LAG) as a source of traffic to be mirrored. The **no** form of this command ceases mirroring traffic from the specified source interface and removes the source interface from the mirroring session configuration.

Parameter	Description
<PORT-NUM>	Specifies a physical port on the switch. Use the format member/slot/port (for example, 1/3/1).
<LAG-NAME>	Specifies the identifier for the LAG (link aggregation group).
<DIRECTION>	Selects the direction of traffic to be mirrored from this source interface. There is no default for this parameter. Valid values are the following:
both	Mirror both transmitted and received packets.
rx	Mirror only received packets.
tx	Mirror only transmitted packets.

Usage

There is a limit of source interfaces in each direction of a given mirror session:

Switch	Source interface limit per mirror session (4 possible sessions)
8400	256
9300	128

However, there is a practical limit to the amount of traffic that a mirror destination can transmit. For example, mirroring session with multiple 10G sources can overwhelm a single 10G destination.

You can configure the same source interface in multiple mirroring sessions, but only one of those mirroring sessions can be enabled at a time.

Classifier policies with mirror actions can also be used to match and mirror network traffic. Although mirror actions of classifier policies must specify an enabled mirroring session, the traffic matching and mirroring actions are separate from and take priority over basic mirroring sessions. For example, mirroring session 1 might monitor a source interface, but a classifier policy might match some traffic from that same source interface and direct it to the destination interface of a different mirroring

session. In this situation, only the traffic that is not matched by the policy is considered for matching by mirroring session 1.

If an interface is in active use by the sFlow feature, then that interface cannot be used as source of received traffic (configured as a source destination with a direction of **rx** or **both**) in an enabled mirroring session. If you want to use this interface as a source of received traffic in a mirroring session, you must disable sFlow on the interface before you enable the mirroring session on the same interface.



When adding, removing, or changing the configuration of a source port in an enabled mirroring session, packets from other mirror sources using the same destination port might be interrupted.

Examples

Configuring a mirrored traffic source interface:

```
switch(config-mirror-1) # source interface  
LAG-NAME      Enter a LAG name. For example, lag10  
PORT-NUM      Enter a port number
```

Creating a mirroring session and configuring a source interface to mirror both transmitted and received packets:

```
switch(config) # mirror session 1  
switch(config-mirror-1) # source interface 1/1/1 both
```

Creating a second mirroring session and configuring two source interfaces. One port mirroring only transmitted packets and the other mirroring both transmitted and received packets:

```
switch(config) # mirror session 2  
switch(config-mirror-2) # source interface 1/1/3 tx  
switch(config-mirror-2) # source interface 1/2/1 both
```

Removing the first source interface:

```
switch(config-mirror-2) # no source interface 1/2/3
```

Configuring a source interface to mirror received packets only:

```
switch(config-mirror-3) # source interface 1/1/2 rx
```

Configuring a source interface to mirror both transmitted and received packets:

```
switch(config-mirror-1) # source interface 1/1/1 both
```

Configuring a LAG as source interface to mirror both transmitted and received packets:

```
switch(config-mirror-4) # source interface lag1 both
```

Stopping the mirroring of received packets from a configured source interface:

```
switch(config-mirror-4) # no source interface lag1 rx
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-mirror-<SESSION-ID>	Administrators or local user group members with execution rights for this command.

source vlan

```
source vlan <VLAN-NUM> {rx | tx | both}
no source vlan <VLAN-NUM> {rx | tx | both}
```

Description

Mirroring with VLAN as a source is supported in the following traffic directions:

- **both** - traffic received and transmitted
- **rx** - only received traffic
- **tx** - only transmitted traffic

More than one source VLAN can be configured in a mirror session. Each such VLAN may specify its own direction.

There is a limit of 6 source VLANs for a given mirror session. There is also a limit of 6 source VLANs across all mirror sessions.

Same VLAN may not be configured as a mirror source for multiple sessions.



When changing a source VLAN in an enabled mirror session (i.e. adding, changing direction, or removing) mirrored packets being transmitted out of the mirror destination port from other mirror sources may be briefly interrupted during the reconfiguration.

Direction of an existing source VLAN can be updated in one of two ways.

- Reenter the **source vlan <VLAN-NUM> <direction>** command with the new preferred direction.
- Use the **no source vlan <VLAN-NUM> <direction>** form of the command with a direction (**rx** or **tx**) to selectively remove the specified direction.

Specifying the last remaining direction for that VLAN will remove the VLAN from the configuration entirely.

Mirroring allows configuration of VLAN as a source. When VLAN source is configured in the **rx** direction, all packets are mirrored as they are received in the switch. When VLAN source is configured in **tx** direction, all packets are mirrored as they are transmitted out of the switch.

For packets bridged through the switch:

- If the mirror is configured in 'both' direction, two copies of packets are mirrored, otherwise one copy of the packet will be mirrored.

For routed packets:

- If the mirror is configured in **rx** direction, packets are mirrored in the pre-routed form with the Destination MAC address as the switch address.

Also, bridged packets are the only packets mirrored in the **rx** direction.

- If the mirror is configured in **tx** direction, packets are mirrored in post-routed form with the source MAC as the switch address. Destination MAC is the nexthop gateway or station.
- If the mirror is configured in **both** direction, one copy of the packet will be mirrored.
- To mirror routed packets received on a VLAN and transmitted out a different VLAN, enable **tx** mirroring on the destination VLAN.

Control plane packets generated by the switch's CPU are processed both in the ingress and the egress packet processing pipeline. The following are the behavior for mirroring with VLAN as source:

- If the mirror is configured in the **rx** or **tx** direction, the packets are mirrored to the mirror destination.
- If the mirror is configured in the **both** direction, two copies of the packets are mirrored to the mirror destination.

The **no** form command will cease mirroring traffic from the specified source VLAN and remove the source from the mirror configuration.

Parameter	Description
VLAN-NUM	Selects the VLAN number.
direction	Specifies the direction of mirroring. tx (transmit), rx (receive), or both .

Examples

Creating a mirror session and adding a VLAN as a source of traffic in both directions on that port:

```
switch# configure terminal
switch(config)# mirror session 1
switch(config-mirror-1)# source vlan 10 both
```

Creating a mirror session and adding two VLANs as sources of traffic:
directions:

```
switch# configure terminal
switch(config)# mirror session 2
switch(config-mirror-2)# source vlan 10 tx
switch(config-mirror-2)# source vlan 20 both
```

Configuring the source in session 2 to receive by specifying the source interface configuration:

```
switch(config-mirror-2) # source vlan 10 rx
```

Removing the first source interface in session 2 entirely, and removing the transmit direction from the other so that mirroring only occurs in the receive direction:

```
switch(config-mirror-2) # source vlan 10 rx  
switch(config-mirror-2) # source vlan 20 tx
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	config	Administrators or local user group members with execution rights for this command.

Monitoring a device using SNMP

Configuring SNMP: Refer to the *SNMP/MIB Guide* for information on how to add SNMP so a device can be monitored from a network management system (NMS).

Configuring an SNMP trap receiver: Refer to the *SNMP/MIB Guide* and specific information about the `show snmp trap` command to enable SNMP traps.

Ports default to an unsplit state. When a port is 'split', the split interfaces become active and can be configured independently. For example, when a 40G QSFP+ port is split four ways, each split interface behaves like a separate 10G SFP+ port. The split interfaces have the same name as the base port with an added suffix to represent their lane of the breakout cable or optical channel on SR4 optics. Splitting an interface removes most of the port's configuration settings and makes it inactive. The port will no longer appear in many show interface commands and most configuration commands are not allowed; the split interface name must be used.

The same thing happens in reverse when an interface is unsplit. However, note that the 'split' and 'no split' commands are always performed in the unsplit port's context.

Limitations with breakout cable support

- The 8400 switch does not support DAC breakout cables, only optical breakout cables.
- The JL365A Aruba 8400X 8p 40G QSFP+ Adv module does not support Priority-Based Flow Control (PFC) on split ports.
- The JL366A Aruba 8400X 6p 40G/100G QSFP28 Adv module does not support 100G breakout cables; it only supports split ports at the 40G speed (into 4x10G links).

Breakout cable support commands

split

```
split [<COUNT>] [<SPEED>] [confirm]
no split [confirm]
```

Description

Splits a port into multiple interfaces. Only ports capable of supporting breakout cables or SR4/eSR4/eDR4 optics can be split.

Parameter	Description
<COUNT>	Specifies the number of child interfaces to activate upon splitting the port. Default: 4.
<SPEED>	Specifies the speed for the child interfaces.
confirm	Specifies the confirmation of port splitting.

Usage

- Some switch interfaces support different split counts depending on the installed transceiver. For these interfaces, the number of child interfaces to activate can be specified. If omitted, the default is

4. For transceivers capable of supporting multiple split modes, the closest mode with enough lanes is used.

- Some transceivers also support multiple split modes with different speeds. For example, 2x200G or 2x100G. When a speed is not specified, the highest available speed for the split count is used. To select a different split mode with a lower speed, the desired speed must be specified.

The splittable ports for all models are shown in the table below:

Model	Description	Port info
8400X modules ■ JL365A ■ JL366A	8400X 8p 40G QSFP+ Adv Module 8400X 6p 40G/100G QSFP28 Adv Module	1-8 (40G) 1-6 Only capable of 40G split into 4 x 10G JL366A modules do not have 25G MACs to support split 100G
8400X modules ■ JL365A ■ JL366A	8400X 8p 40G QSFP+ Adv Mod 8400X 6p 40G/100G QSFP28 Adv Mod	1-8 (40G) 1-6 Only capable of 40G split into 4 x 10G JL366A modules do not have 25G MACs to support split 100G

Examples

Splitting an interface:

```
switch(config-if)# interface 1/1/1
switch(config-if)# split
```

This command will disable the specified port, clear its configuration, and split it into multiple interfaces. The split interfaces will not be available until the next system or line module reboot.

Continue (y/n)? y

```
switch(config-if)# show interface brief
```

```
-----
--
Port      Native Mode   Type           Enabled Status Reason              Speed
Desc      VLAN
-----
--
1/1/1:1 --      routed QSFP+DA3x4  yes          down  Split reboot pending  --  --
1/1/1:2 --      routed QSFP+DA3x4  yes          down  Split reboot pending  --  --
1/1/1:3 --      routed QSFP+DA3x4  yes          down  Split reboot pending  --  --
1/1/1:4 --      routed QSFP+DA3x4  yes          down  Split reboot pending  --  --
```

Unsplitting a port on a switch that requires a reboot:

```
switch(config)# interface 1/1/1
switch(config-if)# no split
```

This command will disable the split interfaces for this port and clear their configuration. The port will not be available until the next system or line module reboot.

Continue (y/n)? y

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	config-if	Administrators or local user group members with execution rights for this command.

You can manage and monitor the AOS-CX switch through Aruba AirWave. The following benefits and functions include:

- Configuration (partial configuration)
- Device topology
- Immediate and historical trend reports
- Monitoring of the device and user connected to the network.
- Network discovery
- Syslogs and trap receiver

For information about which versions of Aruba AirWave support AOS-CX, see the *AOS-CX Release Notes*.

SNMP support and AirWave

For AirWave to discover and monitor the switch, you must:

- Enable the SNMP services on the switch.
- Configure the SNMP agent to use the SNMP version supported by the management station.

SNMP on the switch

The switch provides SNMP services through the management channel and the data interfaces. Functionality, such as device discovery from NMS, syslog and trap forwarding, can be any channel configured by you.

Although the SNMP server can be enabled on both VRFs (`mgmt` and `default`), only one instance of SNMP can be running. The highest priority is on the `default` VRF.

For example, assume that SNMP is first enabled on the `mgmt` VRF (`snmp-server vrf mgmt`). Then, SNMP is enabled on the `default` VRF (`snmp-server vrf default`) without disabling SNMP on the `mgmt` (using an equivalent `no` form of the command). The `show running-config` command displays both `snmp-server vrf` commands; however, the SNMP instance is running only on the `default` VRF (highest priority).

```
switch# config
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server vrf default
switch(config)# show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.01.
led locator on
!
!
!
snmp-server vrf default
```

```
snmp-server vrf mgmt
!
...
```

Supported features with AirWave and the AOS-CX switch

AirWave supports the following features with the AOS-CX switch:

Device management	Device discovery using SNMPv2C and SNMPv3
	Device dashboards
Monitoring management	Device health attributes (device status/reachability)
	Interface and VLAN management
	Initiates an SSH connection from Aruba AirWave to AOS-CX so that the device outputs from the AOS-CX CLI can be displayed in the Aruba AirWave user interface.
	Firmware versions
	Displays neighbor devices connected to AOS-CX switches
Configuration management	Device topology
	Partial configuration
Alarm management	Alarm triggers (device and interface up/down, new device discoveries, custom event triggers)
	Syslogs and traps
Report management	Device inventory, interface utilization, and device reachability reports
	Summary report of device model, firmware, and boot loader version

Configuring the AOS-CX switch to be monitored by AirWave

Prerequisites

Aruba AirWave is active on the network.

Procedure

1. Enable SNMP on the switch by entering the `snmp-server vrf mgmt` command.

```
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server vrf default
```

2. Configure the SNMPv2C community to public by entering the `snmp-server community public` command. In this instance, `public` is a read-only community string.

```
switch(config)# snmp-server community public
```

3. The community-string is used by SNMPv1 and SNMPv2C for unencrypted authentication. SNMPv3 lets you encrypt the authentication mechanism. To enable SNMPv3, enter the `snmpv3 user` and `snmpv3 context` commands.

```
switch(config)# snmpv3 user Admin auth sha auth-pass ciphertext
AQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqbNImqtfYbJYCgAAALkGFJVcSp3nZ3o=
priv des priv-pass ciphertext
AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=

switch(config)# snmpv3 context Admin
```

For discovering devices in AirWave through the SNMPv3 community, the SNMPv3 context name is not mandatory. Devices can still be discovered in Aruba AirWave without the SNMPv3 context name.

4. Enter the `logging` command for enabling syslog forwarding to a remote syslog server, such as AirWave:

```
switch(config)# logging 10.0.10.2 severity debug
```

5. SNMP traps enable an agent to notify the management station of significant events by way of an unsolicited SNMP message. Enable SNMP traps by entering the `snmp-server host` command:

```
switch(config)# snmp-server host 10.10.10.10 trap version v2c vrf default
```

SNMP traps cannot be forwarded from AOS-CX 10.00 switches that have the VRF configured as `mgmt`. Later versions of AOS-CX support SNMP trap forwarding even when the VRF is configured as `default` or `mgmt`.

6. For information on how to add a device for monitoring in the Aruba AirWave user interface, see the documentation for Aruba AirWave.

AirWave commands

logging

```
logging {<IPV4-ADDR> | <IPV6-ADDR> | <FQDN | HOSTNAME>} [ {udp [<PORT-NUM>] }|{tcp
[<PORT-NUM>] | {tls [<PORT-NUM> [auth-mode {certificate|subject-name}] [legacy-tls-
renegotiation]]} [severity <LEVEL>] [vrf <VRF-NAME>] [include-auditable-events]
[filter <FILTER-NAME>] [ rate-limit-burst <BURST> [rate-limit-interval <INTERVAL>] ]
```

```
no logging {<IPV4-ADDR> | <IPV6-ADDR> | <HOSTNAME>}
```

Description

Enables syslog forwarding to a remote syslog server.

The `no` form of this command disables syslog forwarding to a remote syslog server.

Parameter	Description
<code>{<IPv4-ADDR> <IPv6-ADDR> <HOSTNAME>}</code>	Selects the IPv4 address, IPv6 address, or host name of the remote syslog server. Required.
<code>[udp [<PORT-NUM>] tcp [<PORT-NUM>]]</code>	Specifies the UDP port or TCP port of the remote syslog server to receive the forwarded syslog messages.
<code>udp [<PORT-NUM>]</code>	Range: 1 to 65535. Default: 514
<code>tcp [<PORT-NUM>]</code>	Range: 1 to 65535. Default: 1470
<code>tls [<PORT-NUM>]</code>	Range: 1 to 65535. Default: 6514
<code>include-auditable-events</code>	Specifies that auditable messages are also logged to the remote syslog server.
<code>severity <LEVEL></code>	Specifies the severity of the syslog messages: ▪ <code>alert</code>: Forwards syslog messages with the severity of <code>alert</code> (6) and <code>emergency</code> (7). ▪ <code>crit</code>: Forwards syslog messages with the severity of <code>critical</code> (5) and above. ▪ <code>debug</code>: Forwards syslog messages with the severity of <code>debug</code> (0) and above. ▪ <code>emerg</code>: Forwards syslog messages with the severity of <code>emergency</code> (7) only. ▪ <code>err</code>: Forwards syslog messages with the severity of <code>err</code> (4) and above ▪ <code>info</code>: Forwards syslog messages with the severity of <code>info</code> (1) and above. Default. ▪ <code>notice</code>: Forwards syslog messages with the severity of <code>notice</code> (2) and above. ▪ <code>warning</code>: Forwards syslog messages with the severity of <code>warning</code> (3) and above.
<code>auth-mode</code>	Specifies the TLS authentication mode used to validate the certificate. ▪ <code>certificate</code>: Validates the peer using trust anchor certificate based authentication. Default. ▪ <code>subject-name</code>: Validates the peer using trust anchor certificates as well as subject-name based authentication.
<code>legacy-tls-renegotiation</code>	Enables the TLS connection with a remote syslog server supporting legacy renegotiation.
<code>vrf <VRF-NAME></code>	Specifies the VRF used to connect to the syslog server. Optional. Default: <code>default</code>

Examples

Enabling the syslog forwarding to remote syslog server 10.0.10.2:

```
switch(config)# logging 10.0.10.2
```

Enabling the syslog forwarding of messages with a severity of `err` (4) and above to TCP port 4242 on remote syslog server 10.0.10.9 with VRF `lab_vrf`:

```
switch(config)# logging 10.0.10.9 tcp 4242 severity err vrf lab_vrf
```

Disabling syslog forwarding to a remote syslog server:

```
switch(config)# no logging
```

Enabling syslog forwarding over TLS to a remote syslog server using subject-name authentication mode:

```
switch(config)# logging example.com tls auth-mode subject name
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server community

```
snmp-server community <STRING>  
no snmp-server community <STRING>
```

Description

Adds an SNMPv1/SNMPv2c community string. A community string is a password that controls read access to the SNMP agent. A network management program must supply this name when attempting to get SNMP information from the switch. A maximum of 10 community strings are supported. Once you create your own community string, the default community string (`public`) is deleted.

The `no` form of this command removes the specified SNMPv1/SNMPv2c community string. When no community string exists, a default community string with the value `public` is automatically defined.

Parameter	Description
<STRING>	Specifies the SNMPv1/SNMPv2c community string. Range: 1 to 32 printable ASCII characters, excluding space and question mark.

Examples

Setting the SNMPv1/SNMPv2c community string to **private**:

```
switch(config)# snmp-server community private
```


Removing SNMPv1/SNMPv2c community string **private**:

```
switch(config)# no snmp-server community private
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server host

```
snmp-server host <IPv4-ADDR> trap version <VERSION> [community <STRING>]
[port <UDP-PORT>] [vrf <VRF-NAME>]
no snmp-server host <IPv4-ADDR> trap version <VERSION> [community <STRING>]
[port <UDP-PORT>] [vrf <VRF-NAME>]
snmp-server host <IPv4-ADDR> inform version v2c [community <STRING>]
[port <UDP-PORT>] [vrf <VRF-NAME>]
no snmp-server host <IPv4-ADDR> inform version v2c [community <STRING>]
[port <UDP-PORT>] [vrf <VRF-NAME>]
snmp-server host <IPv4-ADDR> [trap version v3 | inform version v3] user <NAME>
[port <UDP-PORT>] [vrf <VRF-NAME>]
no snmp-server host <IPv4-ADDR> [trap version v3 | inform version v3] user <NAME>
[port <UDP-PORT>] [vrf <VRF-NAME>]
```

Description

Configures a trap/informs receiver to which the SNMP agent can send SNMP v1/v2c/v3 traps or v2c informs. A maximum of 30 SNMP traps/informs receivers can be configured.

The **no** form of this command removes the specified trap/inform receiver.



Configuring **snmpv3 informs** is not supported.

Parameter	Description
<IPv4-ADDR>	Specifies the IP address of a trap receiver in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. You can remove leading zeros. For example, the address 192.169.005.100 becomes 192.168.5.100.
trap version <VERSION>	Specifies the trap notification type for SNMPv1 or v2c. Available options are: v1 or v2c.
inform version v2c	Specifies the inform notification type for SNMPv2c.
trap version v3	Specifies the trap notification type for SNMPv3.

Parameter	Description
user <NAME>	Specifies the SNMPv3 user name to be used in the SNMP trap notifications.
community <STRING>	Specifies the name of the community string to use when sending trap notifications. Range: 1 - 32 printable ASCII characters, excluding space and question mark. Default: public.
<UDP-PORT>	Specifies the UDP port on which notifications are sent. Range: 1 - 65535. Default: 162.
vrf <VRF-NAME>	Specifies the name of the VRF on which to send the notifications.

Examples

```

switch(config)# snmp-server host 10.10.10.10 trap version v1
switch(config)# no snmp-server host 10.10.10.10 trap version v1
switch(config)# snmp-server host 10.10.10.10 trap version v2c community public
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
switch(config)# snmp-server host 10.10.10.10 trap version v2c community public
port 5000
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
port 5000
switch(config)# snmp-server host 10.10.10.10 trap version v2c community public
port 5000 vrf default
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
port 5000 vrf default
switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community
public
switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
port 5000
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community
public port 5000
switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
port 5000 vrf default
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community
public port 5000 vrf default

switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin
switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin port 2000
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin port
2000

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server vrf

```
snmp-server vrf <VRF-NAME>  
no snmp-server vrf <VRF-NAME>
```

Description

Configures the VRF on which the SNMP agent listens for incoming requests. By default, the SNMP agent does not listen on any VRF.

The `no` form of this command stops the SNMP agent from listening for incoming requests on the specified VRF.

Parameter	Description
<code><VRF-NAME></code>	Specifies the VRF on which the SNMP agent listens for incoming requests. The SNMP agent can listen on either the <code>mgmt</code> or <code>default</code> VRF. If configured for both, the SNMP agent listens on <code>default</code> , which has a higher priority.

Example

```
switch(config)# snmp-server vrf default
```

```
switch(config)# no snmp-server vrf default
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

snmpv3 context

```
snmpv3 context <NAME> vrf <VRF-NAME> [community <STRING>]  
no snmpv3 context <NAME> [vrf <VRF-NAME>]
```

Description

Creates an SNMPv3 context on the specified VRF.

The `no` form of this command removes the specified SNMP context.

Parameter	Description
<code><NAME></code>	Specifies the name of the context. Range: 1 to 32 printable ASCII

Parameter	Description
	characters, excluding space and question mark (?).
<code>vrf <VRF-NAME></code>	Specifies the VRF associated with the context. Default: default.
<code>community <STRING></code>	Specifies the SNMP community string associated with the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark. Default: public.

Examples

Creating an SNMPv3 context named **newContext**:

```
switch(config)# snmpv3 context newContext
```

Creating an SNMPv3 context named **newContext** on VRF **myVrf** and with community string **private**.

```
switch(config)# snmpv3 context newContext vrf myVrf community private
```

Removing the SNMPv3 context named **newContext** on VRF **myVrf**:

```
switch(config)# no snmpv3 context newContext vrf myVrf
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmpv3 user

```
snmpv3 user <NAME> [auth <AUTH-PROTOCOL> auth-pass {plaintext | ciphertext}
<AUTH-PWORD> [priv <PRIV-PROTOCOL> priv-pass {plaintext | ciphertext} <PRIV-PWORD>] ]
no snmpv3 user <NAME> [auth <AUTH-PROTOCOL> auth-pass
<AUTH-PWORD> [priv <PRIV-PROTOCOL> priv-pass <PRIV-PWORD>] ]
```

Description

Creates an SNMPv3 user and adds it to an SNMPv3 context.

The `no` form of this command removes the specified SNMPv3 user.

Parameter	Description
<code><NAME></code>	Specifies the SNMPv3 username. Range 1 - 32 printable ASCII characters, excluding space and question mark.
<code>auth <AUTH-PROTOCOL></code>	Specifies the authentication protocol used to validate user logins. Available options are: <code>md5</code> or <code>sha</code> .
<code>auth-pass {plaintext ciphertext} <AUTH-PWORD></code>	Specifies the SNMPv3 user password. Range for <code>plaintext</code> is 8 - 32 printable ASCII characters, excluding space and question mark. Range for <code>ciphertext</code> is 1 - 120 printable ASCII characters. This option is only used when copying user configuration settings between switches. It enables you to duplicate a user's configuration on another switch without having to know their password.
<code>priv <PRIV-PROTOCOL></code>	Specifies the SNMPv3 security protocol (encryption method). Available options are: <code>aes</code> or <code>des</code> .
<code>priv-pass {plaintext ciphertext} <PRIV-PWORD></code>	Specifies the SNMPv3 user privacy passphrase. Range for <code>plaintext</code> is 8 - 32 printable ASCII characters, excluding space and question mark. Range for <code>ciphertext</code> is 1 - 120 printable ASCII characters. This option is only used when copying user configuration settings between switches. It enables you to duplicate a user's configuration on another switch without having to know their password.

Examples

Defining an SNMPv3 user named **Admin** using **sha** authentication with the plaintext password **mypassword** and using **des** security with the plaintext password **myprivpass**:

```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword priv des
priv-pass plaintext myprivpass
```

Removing an SNMPv3 user named **Admin**:

```
switch(config)# no snmpv3 user Admin
```

Defining an SNMPv3 user named **Admin** using **sha** authentication with the plaintext password **mypassword** and using **des** security with the plaintext password **myprivpass**:

```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword priv des
priv-pass plaintext myprivpass
```

Copying an SNMP user from switch 1 to switch 2.

On switch 1, configure a user called **Admin**, then issue the `show running-config` command to display switch configuration settings. The `snmpv3 user` command uses the `ciphertext` option to protect the users's passwords.

```
switch1(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword
priv des priv-pass plaintext myprivpass
switch1(config)# exit
switch1# show running-config
Current configuration:
!
!Version AOS-CX TL.10.00.0003-8017-gdeb0606~dirty
!
!
!
snmpv3 user Admin auth sha auth-pass ciphertext
AQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqbNImqtFYbJYCgAAALkGFJVcSp3nZ3o=
priv des priv-pass ciphertext
AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=
ssh server vrf mgmt
!
!
!
!
interface mgmt
    no shutdown
    ip dhcp
    vlan 1
```

On switch 2, execute the `snmpv3 user` command that was displayed by `show running-config` on switch 1. This creates the user on switch 2 with the same configuration settings.

```
switch1(config)# snmpv3 user Admin auth sha auth-pass
ciphertextAQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqbNImqtFYbJYCgAAALkGFJVcSp3nZ3o=priv
des priv-pass ciphertext
AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	https://www.hpe.com/us/en/networking/hpe-aruba-networking-support-services.html
AOS-CX Switch Software Documentation Portal	https://arubanetworking.hpe.com/techdocs/AOS-CX/help_portal/Content/home.htm
HPE Aruba Networking Support Portal	https://networkingsupport.hpe.com/home
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-650-750-0350 (Backup—Toll Number)
International telephone	https://www.hpe.com/psnow/doc/a50011948enw

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	https://developer.arubanetworks.com/hpe-aruba-networking-aoscx/docs/about
Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
AOS-CX Software Technical Update channel on YouTube.	Videos on new features introduced in this release: https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP-Q_UL3CskS

HPE Aruba Networking Hardware Documentation and Translations Portal	https://arubanetworking.hpe.com/techdocs/hardware/DocumentationPortal/Content/home.htm
HPE Aruba Networking software	https://networkingsupport.hpe.com/downloads
Software licensing and Feature Packs	https://licensemanagement.hpe.com/
End-of-Life information	https://networkingsupport.hpe.com/end-of-life

Accessing Updates

You can access updates from the HPE Aruba Networking Support Portal at <https://networkingsupport.hpe.com>.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://networkingsupport.hpe.com/notifications/subscriptions> (requires an active HPE Aruba Networking Support Portal account to manage subscriptions). Security notices are viewable without an HPE Aruba Networking Support Portal account.

Warranty Information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory Information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

Additional regulatory information

HPE Aruba Networking is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see <https://www.arubanetworks.com/company/about-us/environmental-citizenship/>.

Documentation Feedback

HPE Aruba Networking is committed to providing documentation that meets your needs. To help us improve the documentation, send any errors, suggestions, or comments to Documentation Feedback

(docsfeedback-switching@hpe.com). When submitting your feedback, include the document title, part number, edition, and publication date located on the front cover of the document. For online help content, include the product name, product version, help edition, and publication date located on the legal notices page.